BiDCNet: Bidirectional Dual Cross-attention Network for Tomato Leaf Disease Classification

A Report submitted in partial fulfillment for the Degree of

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Submitted by

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(Established Vide Act No. 25 (2009) of Parliament, Govt. of India.)

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Certificate

This is to certify that the project report entitled **BiDCNet: Bidirectional Dual Cross-attention Network for Tomato Leaf Disease Classification** submitted by **Lavudya Raja** to the Central University of Haryana, in partial fulfillment for the award of the degree of **B. Tech. in Computer Science & Engineering** is a bonafide record of project work carried out by him/her under my/our supervision. The contents of this report, in full or in parts, have not been submitted to any other Institution or University for the award of any degree or diploma.

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Declaration

I declare that this project report titled **BiDCNet: Bidirectional Dual Cross-attention Network for Tomato Leaf Disease Classification** submitted in partial fulfillment of the degree of **B. Tech. in Computer Science & Engineering** is a record of original work carried out by me under the supervision of **Dr. Bharti Rana** and **Dr. Vipin Kumar**, and has not formed the basis for the award of any other degree or diploma, in this or any other Institution or University. In keeping with the ethical practice in reporting scientific information, due acknowledgements have been made wherever the findings of others have been cited.

During the preparation of this work, I have not used any AI-based tool to write any part of this project report. I take full responsibility for the submitted content including similarity.

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I would like to acknowledge that this project was completed entirely by me and not by someone else.

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Abstract

Tomato leaf diseases cause significant crop losses worldwide. While Convolutional Neural Networks (CNNs) excel at local texture features and Vision Transformers (ViTs) capture global context, each has inherent limitations. This work proposes **BiDCNet** (Bidirectional Dual Cross-attention Network), a novel dual-branch hybrid architecture that combines a pretrained ResNet-50 CNN encoder with a from-scratch ViT. The two branches produce 196 spatial tokens each at dimension 384, progressively fused through: (i) Full Bidirectional Cross-Attention for dense global-local alignment, and (ii) Sparse Bidirectional Cross-Attention with learnable top-K scoring ($K = 25$) to focus on disease-discriminative regions.

The model is trained on the Augmented Tomato Leaf Disease dataset (43,109 images, 11 classes). Optuna-based Hyperparameter Optimisation with MedianPruner tunes differential learning rates, weight decay, dropout, label smoothing, and top-K over 15 trials. The final model uses AdamW optimiser, Cosine Annealing scheduling, Automatic Mixed Precision, and 5-View Test-Time Augmentation. BiDCNet achieves **99.03%** test accuracy and **0.9903** macro-F1, demonstrating the effectiveness of bidirectional cross-attention fusion for agricultural disease recognition.

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Chapter 1

Objective

1.1 Primary Objective

Among the hybrid architectures explored for plant pathology recognition, **BiDCNet** (Bidirectional Dual Cross-attention Network) occupies a distinct position: a dual-branch deep learning framework engineered to automate fine-grained tomato leaf disease classification across eleven pathological categories. Fungal, bacterial, and viral infections collectively account for considerable yield reductions in tomato cultivation worldwide [1, 2], and the absence of scalable, accurate detection tools continues to hamper timely field intervention. Two architectural lineages have shaped the landscape of automated plant disease recognition. CNN-based models [3] demonstrate strong sensitivity to local textural patterns yet remain constrained by limited receptive fields that preclude meaningful global reasoning. Vision Transformer architectures [4], by contrast, encode long-range spatial dependencies naturally, but their data-hungry training dynamics render them poorly suited to agricultural datasets of moderate scale. BiDCNet resolves this tension through a dual-stage bidirectional cross-attention mechanism that links a pretrained ResNet-50 backbone with a randomly initialised ViT branch, enabling each sub-network to selectively condition its representations on the other's feature maps without excessive parameter overhead.

All experiments were conducted in PyTorch on the Augmented Tomato Leaf Disease dataset, which contains 43,109 annotated leaf images partitioned into 34,243 training, 4,433 validation, and 4,433 test samples. The intended benchmark for the final model stands at a test accuracy of **99.03%** and a macro-averaged F1 score of **0.9903**.

1.2 Specific Objectives

1. **Dataset Preparation and Preprocessing:** The Augmented Tomato Leaf Disease dataset, spanning 11 balanced pathology classes, was loaded and each image spatially standardised to 224×224 pixels to satisfy the input requirements of both architectural branches. Channel-wise normalisation was performed using ImageNet-derived statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$), ensuring the input distribution remained consis-

tent with the pretrained ResNet-50 backbone's weight calibration. Data augmentation was applied exclusively during training and comprised random horizontal and vertical flipping, rotational perturbation within $\pm 15°$, colour jitter, stochastic greyscale conversion, and random erasing [5]. Validation and test partitions underwent only resizing and normalisation, preserving an unbiased evaluation environment.

2. **Dual-Branch Feature Extraction:** Implement two parallel encoders on the same input image:

   - *CNN Branch (pretrained):* ResNet-50 [3] via `timm`, extracting `layer3` features ($14 \times 14 \times 1024$), projecting to $D = 384$ with 1×1 convolution, Batch Normalisation, and GELU, and forming 196 spatial tokens.
   - *ViT Branch (from scratch):* Lightweight ViT with patch size 16, embedding dimension 384, depth 4, and 6 attention heads, initialised randomly and trained on the tomato disease domain.

3. **Dual-Stage Bidirectional Cross-Attention Fusion:**

   - *Stage 1 — Full Bidirectional Cross-Attention:* Dense multi-head cross-attention between all 196 CNN tokens and all 196 ViT tokens in both directions, with residual connections and feed-forward refinement.
   - *Stage 2 — Sparse Bidirectional Cross-Attention:* Learnable per-token scoring to select top-K discriminative tokens from each branch (implementation uses $K = 49$), followed by sparse bidirectional cross-attention and scatter-back to the full token grid.

4. **Feature Fusion and Classification:** Global-average-pool the refined CNN tokens; combine the ViT CLS token and mean patch representation; concatenate, fuse through a linear layer with Layer Normalisation and dropout, and classify with a two-layer MLP head (hidden size 256) for 11 disease classes.

5. **Training of the Proposed BiDCNet Model:** BiDCNet was trained over a maximum of 60 epochs on the complete training partition. Optimisation employed AdamW [6] with differential learning rates assigned per branch — 2×10^{-5} for the pretrained CNN backbone and 1×10^{-4} for all scratch-initialised modules — alongside a weight decay of 10^{-4}, label smoothing of 0.05, and a dropout rate of 0.1. The learning rate schedule followed Cosine Annealing [7], with gradient clipping enforced at a maximum norm of 1.0 and Automatic Mixed Precision enabled throughout to accelerate computation on an NVIDIA Tesla T4 GPU. Training was subject to early stopping conditioned on validation macro-F1, with a patience threshold of 20 epochs; the checkpoint yielding peak validation performance was retained for downstream evaluation.

6. **Evaluation of BiDCNet:** The retained best checkpoint was assessed against the held-out test partition comprising 4,433 images. Reported

metrics include overall test accuracy, macro-averaged F1 score (target: 0.9903), and per-class precision, recall, and F1 values for each of the 11 disease categories. A full confusion matrix was generated to visualise inter-class discrimination behaviour.

7. **Comparative Benchmark Against 15 State-of-the-Art Models:** Fifteen hybrid architectures drawn from the recent literature were reproduced and evaluated under identical conditions — the same dataset splits, preprocessing pipeline, and Optuna-based hyperparameter optimisation protocol — yielding a consolidated comparison pool of **16 models** inclusive of BiDCNet. The analysis examined parameter counts, training, validation, and test metrics, and confusion-matrix characteristics across all configurations, with the aim of situating the proposed fusion design within the broader performance landscape.

8. **Ablation Study:** The contribution of individual fusion components was isolated through controlled ablation experiments. Four architectural variants were trained and evaluated independently: a ResNet-50-only baseline, a standalone ViT branch, a bidirectional cross-attention configuration, and the complete BiDCNet incorporating sparse Stage 2 attention. Training, validation, and test accuracy were recorded for each variant to quantify the marginal gain attributable to each design decision.

Chapter 2

Introduction

Across tomato-growing regions worldwide, leaf diseases follow a pattern that farmers know all too well: infection spreads silently through a field, and visible symptoms emerge only after the pathogen has already established itself across a significant area. Classified among the most economically consequential vegetable crops globally, tomato remains acutely susceptible to a spectrum of fungal, bacterial, viral, and pest-induced conditions capable of cutting seasonal yields by thirty to eighty percent when detection arrives too late [1, 2]. Conventional responses — dispatching an agronomist for an on-site assessment or forwarding physical leaf samples to a diagnostic laboratory — demand time and resources that smallholder farmers in remote areas rarely possess. For these growers, a week-long wait for expert confirmation is not a minor inconvenience; it is the difference between a manageable outbreak and a lost harvest.

This work is oriented around one practical question: whether a photograph of a diseased tomato leaf, captured on a standard smartphone and passed through a purpose-built deep learning system, can yield a reliable, eleven-class disease identification. The answer this project demonstrates is affirmative — but reaching that outcome required confronting a computer vision problem that current architectures address only incompletely.

2.1 Background and Motivation

2.1.1 The Agricultural Problem

Tomato cultivation occupies millions of hectares across South Asia, East Asia, North America, and Southern Europe, with India, China, the United States, and Turkey accounting for the majority of global output [1]. The crop's economic centrality, however, exists alongside considerable biological vulnerability. At least eleven distinct foliar diseases afflict tomato plants in regular cultivation, each driven by a different pathogen class and each demanding a different agrochemical or management response. Misidentification carries real cost: administering a fungicide against a bacterial infection neither arrests the disease nor spares the plant from the chemical burden, while a viral outbreak mistaken for nutritional stress may go entirely untreated until irreversible damage has occurred.

Field-based detection has historically depended on trained agronomists conducting site visits — a model that functions in intensively managed farms but collapses in under-resourced or geographically isolated settings. A farmer observing unusual lesions on a Tuesday morning cannot restructure her week around a specialist appointment. A photograph taken in that moment, however, carries all the visual information a well-trained classifier needs. Bridging the gap between that image and an actionable diagnosis is the central practical motivation for this project.

2.1.2 Why Deep Learning is the Right Tool

Image-based plant disease detection has a longer history than deep learning itself. Earlier pipelines extracted hand-engineered features — colour histograms, edge maps, Gabor texture descriptors — and fed them into classical classifiers such as support vector machines [8]. Under controlled laboratory illumination and against uniform backgrounds, these systems produced acceptable results. Deployed in real agricultural environments, where leaf orientation, ambient lighting, and image backgrounds vary continuously, their performance degraded significantly.

Deep neural networks resolved this brittleness by learning discriminative features directly from labelled image collections rather than requiring a human to specify them in advance. The publication of the PlantVillage dataset [9], containing over 50,000 annotated healthy and diseased leaf images spanning multiple crop species, provided the training corpus that made reliable deep learning-based plant pathology feasible. Convolutional models trained on these data rapidly achieved, and in several benchmarks exceeded, the diagnostic accuracy of human domain experts on held-out test sets.

The objective here, however, is not simply to add one more classification result to that literature. It is to examine where the best current models still fall short on genuinely ambiguous disease pairs, and to construct an architecture whose design specifically addresses those failure modes.

2.2 Problem Statement

Fine-grained tomato leaf disease classification is substantially more demanding than aggregate accuracy figures on standard benchmarks suggest. Within the eleven-class taxonomy used in this work, several category pairs share overlapping visual signatures at the scale of a leaf photograph. Early Blight and Target Spot both present with concentric ring-shaped necrotic lesions; at moderate image resolution, the two are visually difficult to separate. Bacterial Spot and Septoria Leaf Spot both manifest as dispersed dark spots across the lamina surface, differing primarily in lesion margin characteristics. Spider Mite damage and early healthy leaf states share a comparable pale-green discolouration, particularly under suboptimal photography conditions. A classifier that cannot reliably resolve these pairs will fail precisely when diagnostic precision matters most — and will likely do so with high confidence.

Accurate resolution of these ambiguous cases demands two qualitatively different forms of visual reasoning, applied simultaneously. Lesion identification requires fine-grained local analysis: the geometry of a spot's boundary, the colour gradient between an infected patch and surrounding tissue, and the surface texture of a necrotic region. Contextual interpretation requires global spatial reasoning: the distribution of symptoms across the leaf, their proximity to major veins, and whether the pattern of spread is uniform or clustered. These two requirements sit in tension with each other at the architectural level.

Convolutional Neural Networks (CNNs), exemplified by ResNet-50 [3], are highly effective at local feature extraction. Learned filters applied across the spatial dimensions of an image capture texture, colour, and edge information with considerable precision. The architectural limitation is inherent: a convolutional filter operates within a bounded receptive field, and the global relationships between spatially distant image regions can only be approximated through deep stacking, which introduces feature abstraction that progressively loses spatial resolution.

Vision Transformers (ViTs) [4] invert this trade-off. By decomposing an image into non-overlapping patches and applying multi-head self-attention across the full patch sequence, ViTs compute pairwise spatial relationships globally at every layer. This global inductive bias comes at a cost: ViTs lack the spatial locality priors that give CNNs their efficiency on texture-sensitive tasks, and their convergence without large-scale pre-training is unreliable on datasets of the size typical in agricultural computer vision.

Late fusion — the predominant strategy for combining CNN and ViT representations — addresses neither limitation adequately. Concatenating the two branches' final embedding vectors at the classification stage allows each model to contribute independently, but prohibits either from influencing the other's internal feature construction. The opportunity for mutual representational refinement — CNN attention guided by the ViT's global context, ViT patch selection informed by the CNN's local saliency — is never exploited.

Formally, given a tomato leaf image $\mathbf{x} \in \mathbb{R}^{3\times224\times224}$, the learning objective is to construct a classifier $f_\theta : \mathbf{x} \mapsto \hat{y} \in \{1, \ldots, 11\}$ that correctly assigns one of eleven disease labels. The architectural requirement is that f_θ facilitates active, bidirectional feature exchange between the CNN and ViT branches throughout the representation-building process — not solely at the final pooling stage — and that this exchange is spatially selective, concentrating on the image regions that carry the greatest diagnostic information.

2.3 Project Objectives

Four technical objectives structure this investigation.

1. **Design a dual-branch hybrid architecture** in which a pretrained CNN backbone and a scratch-trained Vision Transformer are coupled through bidirectional cross-attention at two intermediate processing stages, rather than merged only at the output.

2. **Implement sparse token selection** at the second cross-attention stage, restricting mutual attention to the top-K most diagnostically relevant spatial positions in each branch and thereby concentrating representational capacity on lesion-bearing regions while reducing computational overhead.

3. **Automate hyperparameter search** via Optuna [10] with Tree-structured Parzen Estimator sampling across fifteen trials, replacing manual grid search with a principled optimisation procedure and ensuring that reported results correspond to a properly calibrated configuration.

4. **Conduct rigorous comparative evaluation** of the proposed model against nineteen baseline architectures trained under identical dataset partitions and preprocessing conditions, reporting accuracy, macro-F1, per-class precision, recall, and confusion matrices augmented with Test-Time Augmentation.

2.4 Scope and Limitations

This project addresses tomato leaf disease classification on the Augmented PlantVillage dataset, a corpus of 43,109 images distributed across eleven pathology categories under controlled photographic conditions. The investigation encompasses the complete pipeline from dataset ingestion and augmentation through architecture design, training, Optuna-based hyperparameter search, and held-out test evaluation.

Four constraints bound the scope of this work. The dataset consists of images captured against uniform backgrounds; no evaluation on field photographs involving soil occlusion, partial leaves, or variable ambient illumination has been conducted, and generalisation to such conditions cannot be assumed. The classification taxonomy is restricted to eleven tomato-specific disease categories and does not extend to other solanaceous crops or to pathologies outside this set. Optuna-driven optimisation covered key training hyperparameters, but the core architectural decisions — Transformer depth, token dimensionality, and the dual-stage cross-attention topology — were fixed a priori by design rationale rather than included in the search space. Finally, inference was conducted on GPU hardware; the model has not been quantised or otherwise adapted for deployment on mobile or edge devices, which remains an essential step before practical field deployment.

2.5 Overview of Methodology

BiDCNet (Bidirectional Dual Cross-attention Network) routes each input leaf image through two parallel feature extraction branches. A ResNet-50 backbone pretrained on ImageNet [3] constitutes the CNN branch, converting the 224×224 input into 196 spatial tokens that encode locally learned texture and colour representations. A lightweight Vision Transformer [4] initialised from random weights forms the second branch, producing 196 patch tokens that model spatial co-occurrence relationships across the full leaf surface.

Inter-branch communication occurs at two successive stages. The first stage applies full bidirectional cross-attention [11]: every CNN token attends over the complete set of ViT tokens, and every ViT token simultaneously attends over the complete set of CNN tokens, ensuring unrestricted mutual information transfer before any spatial selection is applied. The second stage narrows this exchange through learned scoring: a lightweight network scores each token in both branches and restricts the subsequent cross-attention round to the top-K highest-scoring positions, directing representational capacity toward the spatial locations most indicative of disease pathology.

Terminal aggregation combines average-pooled CNN tokens with the ViT branch's CLS token and patch mean into a single concatenated vector, which a two-layer projection head maps to class probabilities over the eleven disease categories. The optimiser is AdamW [6] with branch-differentiated learning rates — lower for the pretrained backbone, higher for scratch-trained components — governed by Cosine Annealing scheduling [7] and supplemented by label smoothing. Hyperparameter selection is delegated to Optuna over fifteen trials. At inference, predictions from five geometric augmentations of each test image are averaged to produce the final classification decision.

On 4,433 held-out test images, BiDCNet achieves a test accuracy of **99.03%** and a macro-F1 of **0.9903**, placing it at the top of the comparative evaluation pool.

2.6 Organisation of This Report

The chapters that follow are structured as below.

- **Chapter 3 (Basic Concepts):** Covers the foundational deep learning components underlying BiDCNet — convolutional networks, Vision Transformers, attention mechanisms, and regularisation strategies.
- **Chapter 4 (Literature Survey):** Examines prior CNN-based, ViT-based, and hybrid approaches to plant disease classification and identifies the representational gap that motivates the proposed architecture.
- **Chapter 5 (Problem Definition and Proposed Work):** States the research problem in formal terms and details the BiDCNet architecture, training protocol, and primary contributions.
- **Chapter 6 (Methodology):** Presents the complete system design including mathematical formulations for the cross-attention mechanism, the dataset pipeline, augmentation strategy, and Optuna search configuration.
- **Chapter 7 (Results and Discussion):** Reports experimental outcomes — per-class performance, comparative rankings, confusion-matrix analysis, and an candid treatment of failure cases.
- **Chapter 8 (Conclusion):** Synthesises the principal findings, assesses what the results establish and where uncertainty remains, and proposes concrete directions for subsequent investigation.

Chapter 3

Basic Concepts

The architectural decisions underlying BiDCNet draw on several foundational ideas in deep learning. Each concept introduced in this chapter is first described intuitively, then stated in formal terms. Familiarity with these building blocks is necessary to understand how BiDCNet's CNN and Vision Transformer branches are coupled and why that coupling improves disease classification performance.

3.1 Convolutional Neural Networks (CNNs)

3.1.1 What is a CNN?

Convolutional Neural Networks are deep learning architectures constructed specifically for data arranged in a spatial grid images being the most common case. Their defining operation applies a small learnable filter, termed a *kernel*, across the entire spatial extent of an input, producing a response wherever a target pattern is detected regardless of its position. A kernel trained to recognise horizontal edges, for instance, will activate on that pattern whether it appears near the image boundary or at its centre.

3.1.2 How CNNs Work

Three layer types constitute the repeating structure of a CNN:

- **Convolutional Layer:** F filters, each of spatial extent $k \times k$, are translated across the input feature map at a defined stride. At each position, the filter computes an inner product with the local input patch. For input $\mathbf{X} \in \mathbb{R}^{C\times H\times W}$ and filter $\mathbf{W} \in \mathbb{R}^{C\times k\times k}$, the output at position (i, j) is:

$$(\mathbf{X} * \mathbf{W})_{i,j} = \sum_{c=1}^{C}\sum_{m=1}^{k}\sum_{n=1}^{k} \mathbf{X}_{c,\, i+m,\, j+n} \cdot \mathbf{W}_{c,m,n} \tag{3.1}$$

- **Activation Function (ReLU/GELU):** A pointwise nonlinearity applied after each convolution. ReLU is defined as $\mathrm{ReLU}(x) = \max(0, x)$; GELU [12] provides a smoother alternative. Without such nonlinearities, an arbitrarily deep stack of convolutional layers collapses to a single affine transformation, preventing the network from representing complex mappings.

- **Pooling Layer:** Spatial downsampling,for example, 2×2 max-pooling halves both height and width reduces the number of subsequent computations and introduces a degree of spatial invariance to small translations.

3.1.3 ResNet-50 and Skip Connections

ResNet-50 [3] is a 50-layer architecture whose central innovation is the *residual connection.* Rather than fitting a direct mapping $\mathbf{y} = \mathcal{F}(\mathbf{x})$, each block is reformulated to learn the residual quantity:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}) + \mathbf{x} \tag{3.2}$$

The practical consequence is substantial: gradients propagate directly from later layers back to earlier ones without being attenuated through repeated nonlinear transformations, making reliable optimisation of very deep networks feasible. ResNet-50 attains approximately 76% top-1 accuracy on ImageNet and serves as the CNN backbone within BiDCNet.

3.1.4 Limitation of CNNs

Convolutional filters operate over a spatially bounded region of size $k \times k$. Expanding the effective receptive field to encompass the full image requires stacking many layers in succession. Even so, no single operation directly relates a lesion appearing in one quadrant of a leaf to a discolouration pattern in the opposite quadrant. This absence of a direct global relational mechanism constitutes the **global context limitation** of convolutional architectures.

3.2 Transfer Learning and ImageNet Pretraining

3.2.1 What is Transfer Learning?

Training a large neural network from random initialisation demands millions of labelled samples and substantial GPU time. Transfer learning circumvents this requirement by beginning from a model already optimised on a large general-purpose dataset, then adapting its parameters to the target task through continued training.

3.2.2 ImageNet Pretraining

ImageNet-1K comprises 1.28 million images distributed across 1,000 object categories. Networks trained on this corpus develop general-purpose visual representations in their early layers — edge detectors, texture filters, colour-selective units, and shape-sensitive features — while deeper layers encode higher-level semantic structure. These representations transfer productively to downstream tasks, including agricultural disease recognition, even when the visual domain differs considerably from the pretraining data.

Within BiDCNet, the ResNet-50 backbone is loaded with ImageNet-pretrained weights, supplying the model with established texture and shape detectors before any tomato leaf image is presented. Continued optimisation at a reduced learning rate of 4.79×10^{-5} preserves the accumulated general knowledge while allowing disease-specific adaptation to proceed.

3.3 Self-Attention and Transformers

3.3.1 The Attention Mechanism

The attention mechanism [11] allows a model to dynamically weight contributions from different positions in an input sequence when constructing each output element. Three derived matrices drive the computation:

1. A **Query** $\mathbf{Q}$: encodes what each token is seeking.
2. A **Key** $\mathbf{K}$: encodes what each token exposes to others.
3. A **Value** $\mathbf{V}$: holds the content to be selectively aggregated.

Scaled dot-product attention computes a weighted sum of the value vectors, with weights determined by query-key compatibility:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\top}}{\sqrt{d_k}}\right)\mathbf{V} \tag{3.3}$$

The scaling factor $\sqrt{d_k}$ prevents the dot products from growing large enough to push the softmax into regions of near-zero gradient. The output at position i is a weighted combination of all value vectors, where the weight on position j reflects how relevant j is to i.

3.3.2 Multi-Head Self-Attention (MHSA)

Rather than performing a single attention computation, Multi-Head Attention executes H parallel attention operations, each operating on independently learned linear projections of $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$. The outputs of all heads are concatenated and linearly mixed:

$$\begin{aligned}\text{MHSA}(\mathbf{X}) &= \text{Concat}(\text{head}_1, \ldots, \text{head}_H)\,\mathbf{W}^O, \\ \text{head}_h &= \text{Attention}(\mathbf{X}\mathbf{W}_h^Q,\, \mathbf{X}\mathbf{W}_h^K,\, \mathbf{X}\mathbf{W}_h^V)\end{aligned} \tag{3.4}$$

Distributing attention across multiple heads permits simultaneous modelling of distinct relational patterns — one head may specialise in spatially proximate dependencies while another captures relationships between distant regions. In BiDCNet, both the ViT self-attention layers and the cross-attention fusion modules operate with $H = 6$ heads over $D = 384$-dimensional tokens, assigning $d_k = 64$ dimensions per head.

3.3.3 Transformer Block

Each Transformer block applies MHSA followed by a position-wise Feed-Forward Network (FFN), with residual connections and Layer Normalisation preceding each sub-layer (Pre-LN design):

$$\mathbf{Z}' = \mathbf{Z} + \mathrm{MHSA}(\mathrm{LN}(\mathbf{Z})) \tag{3.5}$$

$$\mathbf{Z}_{\mathrm{out}} = \mathbf{Z}' + \mathrm{FFN}(\mathrm{LN}(\mathbf{Z}')) \tag{3.6}$$

The FFN applies two linear projections with an intermediate GELU activation: $\mathrm{FFN}(\mathbf{z}) = \mathbf{W}_2\,\mathrm{GELU}(\mathbf{W}_1\,\mathbf{z})$, where $\mathbf{W}_1 \in \mathbb{R}^{D\times 4D}$ and $\mathbf{W}_2 \in \mathbb{R}^{4D\times D}$.

3.4 Vision Transformer (ViT)

3.4.1 From NLP to Images

Transformer architectures were originally developed for sequential text, where tokens correspond to words or sub-word units. Dosovitskiy et al. [4] extended this framework to visual inputs by reformulating an image as an ordered sequence of patch tokens:

1. A 224×224 image is partitioned into non-overlapping 16×16 patches, producing $\frac{224}{16} \times \frac{224}{16} = 14 \times 14 = 196$ patches in total.
2. Each patch is flattened from its raw $16\times16\times3 = 768$ values and projected to a D-dimensional vector via a learned linear layer, yielding the *patch embedding*.
3. A learnable `[CLS]` token is prepended to the sequence, tasked with aggregating a global image representation.
4. Learnable *positional embeddings* are added element-wise to inject spatial ordering, compensating for the permutation-invariance of self-attention.
5. The resulting $N+1$ token sequence is processed through L stacked Transformer blocks.
6. The final `[CLS]` token state is passed to the classification head.

3.4.2 Advantage over CNNs

Self-attention evaluates pairwise relationships across the complete token set within each layer. A patch encoding a lesion in the upper-left region of the leaf can therefore directly modulate the representation of a patch in the lower-right region through a single attention operation, without relying on the incremental receptive field expansion that deep CNNs require. This constitutes a genuine architectural advantage for tasks requiring whole-image contextual reasoning.

3.4.3 Limitation

Attention computation scales as $O(N^2)$ in the number of tokens. At $N = 196$ with $H = 6$ heads the cost remains tractable, but higher input resolutions amplify this expense rapidly. More consequentially for this work, ViTs trained from random initialisation on datasets of moderate size — fewer than several hundred thousand images — frequently underperform comparable CNNs, because they lack the translation-equivariance and local spatial inductive biases that convolutional architectures possess inherently.

3.4.4 CNN vs. Vision Transformer: A Comparison

Table 3.1 summarises the principal architectural differences between CNNs and ViTs. BiDCNet incorporates both branches to exploit the complementary strengths of each.

Table 3.1: CNN vs. Vision Transformer: Key Differences

Feature	**CNN (ResNet-50)**	**Vision Transformer (ViT)**
Feature extraction	Local: small kernel scans the image	Global: all patches attend to each other
Receptive field	Grows slowly with depth	Full image in one attention layer
Training data needed	Works well on small datasets	Needs large datasets from scratch
Spatial inductive bias	Strong (translation equivariance)	Minimal (no built-in spatial prior)
Best strength	Textures, edges, lesion spots	Whole-leaf context and relationships
Attention complexity	$O(N \cdot k^2)$ per layer	$O(N^2)$ per attention layer
Role in BiDCNet	Branch A (pretrained, fine-tuned)	Branch B (trained from scratch)

3.5 Cross-Attention

3.5.1 Self-Attention vs. Cross-Attention

When all three matrices **Q**, **K**, and **V** are derived from the same sequence, the operation is termed **self-attention** — each token attends over its own sequence. **Cross-attention** modifies this arrangement so that queries originate from one sequence while keys and values are drawn from a separate sequence:

$$\text{CrossAttn}(\mathbf{X}_{\text{target}},\, \mathbf{X}_{\text{source}}) = \text{softmax}\left(\frac{(\mathbf{X}_{\text{target}}\mathbf{W}^{Q})\,(\mathbf{X}_{\text{source}}\mathbf{W}^{K})^{\top}}{\sqrt{d_k}}\right)\mathbf{X}_{\text{source}}\mathbf{W}^{V} \tag{3.7}$$

Each target token queries the source sequence and assembles a weighted combination of source values, producing a target representation enriched by information from an external stream.

Table 3.2 contrasts self-attention and cross-attention across their defining properties.

Table 3.2: Self-Attention vs. Cross-Attention

Property	**Self-Attention**	**Cross-Attention**
Query source	Same sequence	Target sequence (e.g., CNN tokens)
Key / Value source	Same sequence	Different sequence (e.g., ViT tokens)
Purpose	Internal token relationships within one branch	Communication between two different branches
Used in BiDCNet	ViT branch (4 Transformer blocks)	Fusion Stage 1 (FCA) and Stage 2 (SCA)
Output	Updated tokens from same branch	Target tokens enriched by source branch info

3.5.2 Bidirectional Cross-Attention

BiDCNet applies cross-attention concurrently in both directions between the two branches:

- CNN → ViT: CNN spatial tokens query the ViT patch sequence, incorporating global contextual cues into locally grounded representations.
- ViT → CNN: ViT patch tokens query the CNN spatial sequence, grounding globally contextual representations in precise local texture information.

Neither branch occupies a passive role in this arrangement. Both continuously refine their internal representations by drawing on the complementary information maintained by the other.

Table 3.3 contrasts unidirectional and bidirectional cross-attention strategies.

3.6 Sparse Attention and Top-K Token Selection

3.6.1 The Motivation for Sparsity

Full cross-attention between two token sequences of length N incurs $O(N^2)$ score computations. At $N = 196$, this amounts to 38,416 operations per attention head. Rather than repeating this complete 196×196 computation at Stage

Table 3.3: Unidirectional vs. Bidirectional Cross-Attention

Property	**Unidirectional**	**Bidirectional (BiDC-Net)**
Direction of flow	CNN $\rightarrow$ ViT only	CNN $\leftrightarrow$ ViT both ways
CNN tokens updated?	No	Yes, using ViT global context
ViT tokens updated?	Yes	Yes, using CNN texture detail
Branch communication	One-way handoff	Active mutual dialogue
Example models	Sequential pipelines (PlantViT)	BiDCNet Stage 1 and Stage 2
Disease region focus	Partial	Both branches focus on lesion areas

2, BiDCNet restricts the second cross-attention round to the K tokens judged most diagnostically relevant in each branch, directly reducing the quadratic cost.

Table 3.4 contrasts the full attention at Stage 1 with the sparse formulation at Stage 2.

Table 3.4: Full Cross-Attention (Stage 1) vs. Sparse Cross-Attention (Stage 2) in BiDCNet

Property	**Stage 1: Full (FCA)**	**Stage 2: Sparse (SCA)**
Tokens involved	All 196 $\times$ 196 tokens	Top-K = 25 tokens per branch
Attention complexity	$O(N^2)$ = 38,416 ops/head	$O(K^2)$ = 625 ops/head
Purpose	Complete global–local alignment	Focus on disease-relevant regions
Token selection	None (all tokens used)	Learnable scorer (LN + Linear)
Computational cost	Higher	$\sim$60$\times$ lower than full
Output	All tokens updated	Only top-K tokens updated

3.6.2 Learnable Token Scoring

A compact scoring network — Layer Normalisation followed by a linear projection to a scalar — evaluates the relevance of each token independently:

$$s_i = \mathbf{w}^\top \mathrm{LN}(\mathbf{t}_i) + b, \quad i = 1, \ldots, 196 \tag{3.8}$$

The K tokens attaining the highest scores are identified:

$$\mathcal{I} = \text{argtop-}K(\{s_1, s_2, \ldots, s_{196}\}) \tag{3.9}$$

Cross-attention at Stage 2 is computed exclusively over these selected tokens. Their updated representations are then scattered back to their original sequence positions, while the remaining tokens retain the representations produced at Stage 1. The value of K was determined through the Optuna HPO study, with $K = 25$ emerging as optimal from the candidate set $\{25, 36, 49, 64\}$.

3.6.3 Why Learnable Scoring is Better

Fixed heuristics such as token L2-norm have previously been used to proxy token importance, but they are calibrated to no particular task. A learnable scoring network offers three concrete advantages: it is optimised jointly with the full model through end-to-end gradient flow; it acquires the capacity to assign elevated scores to tokens overlying lesion centres, spots, and necrotic margins while suppressing background tokens; and its decisions reflect the specific discriminative demands of eleven-class tomato pathology recognition rather than any task-agnostic magnitude measure.

3.7 Normalisation Techniques

3.7.1 Batch Normalisation (BN)

Batch Normalisation standardises convolutional layer outputs across the batch dimension. Given a mini-batch of B samples, each channel's activations are shifted to zero mean and unit variance, then rescaled by learnable affine parameters γ and β:

$$\hat{x}_i = \frac{x_i - \mu_{\mathcal{B}}}{\sqrt{\sigma_{\mathcal{B}}^2 + \epsilon}}, \quad y_i = \gamma \hat{x}_i + \beta \tag{3.10}$$

This normalisation stabilises gradient flow and accelerates CNN convergence [13]. Within BiDCNet, it appears in the CNN projection head following a 1×1 convolution and preceding the GELU activation.

3.7.2 Layer Normalisation (LN)

Layer Normalisation operates across the embedding dimension D for each individual token, independently of batch composition:

$$\mathrm{LN}(\mathbf{x}) = \frac{\mathbf{x} - \mu}{\sqrt{\sigma^2 + \epsilon}} \cdot \boldsymbol{\gamma} + \boldsymbol{\beta}, \quad \mu = \frac{1}{D}\sum_{d=1}^{D} x_d, \quad \sigma^2 = \frac{1}{D}\sum_{d=1}^{D}(x_d - \mu)^2 \tag{3.11}$$

Because its statistics are computed per token rather than per batch, LN remains well-defined even when batch size equals one — a property that makes it the standard normalisation choice for Transformer architectures [14]. All ViT blocks, cross-attention modules, and the fusion projection layer in BiDCNet employ LN.

3.8 Dropout and Label Smoothing (Regularisation)

3.8.1 Dropout

During training, dropout stochastically zeroes a fraction p of each layer's activations, preventing individual neurons from developing strong mutual dependencies on specific counterparts elsewhere in the network. The effect approximates averaging predictions across an exponentially large ensemble of subnetworks. All activations are restored at inference time, with outputs scaled to maintain the expected magnitude. Dropout is applied within all FFN sub-layers, the fusion projection, and the classification head in BiDCNet; the rate $p = 0.122$ was selected automatically by the Optuna HPO study.

3.8.2 Label Smoothing

Conventional cross-entropy training drives the predicted probability of the correct class toward 1.0 and all others toward 0.0, a regime that systematically produces overconfident models. Label smoothing [15] redistributes this target mass, replacing the hard one-hot vector with a soft distribution:

$$q_c = \begin{cases} 1 - \varepsilon & \text{if } c = \text{correct class} \\ \dfrac{\varepsilon}{C-1} & \text{otherwise} \end{cases} \tag{3.12}$$

where ε denotes the smoothing coefficient and $C = 11$ is the number of disease categories. At the HPO-selected value of $\varepsilon = 0.065$, the correct class target is reduced from 1.0 to 0.935, and each incorrect class receives $0.065/10 = 0.0065$, moderating the model's confidence on training examples and improving held-out generalisation.

3.9 Optimiser: AdamW and Cosine Annealing

3.9.1 AdamW Optimiser

AdamW [6] maintains running estimates of the first and second moments of each parameter's gradient, using these to construct an adaptive per-parameter learning rate:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t \qquad \text{(first moment: gradient direction)} \tag{3.13}$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \qquad \text{(second moment: gradient magnitude)} \tag{3.14}$$

$$\theta_t = \theta_{t-1} - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} - \alpha\lambda\theta_{t-1} \tag{3.15}$$

The terminal term $-\alpha\lambda\theta$ applies L2 weight decay directly to the parameter vector, decoupled from the adaptive gradient scaling — a theoretically sounder formulation than the standard Adam implementation. BiDCNet employs differential learning rates under AdamW: the pretrained ResNet-50 backbone receives a lower rate to protect its accumulated ImageNet representations, while all scratch-trained components are updated at a higher rate to support faster task-specific adaptation.

3.9.2 Cosine Annealing LR Scheduler

The learning rate decays from its initial value to a minimum of $\eta_{\min} = 10^{-6}$ along a cosine trajectory spanning all 60 training epochs:

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min})\left(1 + \cos\left(\frac{t}{T_{\max}}\pi\right)\right) \tag{3.16}$$

The smooth, monotonically decreasing schedule [7] avoids the abrupt learning rate drops characteristic of step-decay policies, allowing the optimiser to settle progressively into a well-converged minimum.

3.10 Hyperparameter Optimisation with Optuna

3.10.1 What is a Hyperparameter?

Hyperparameters govern aspects of model training that lie outside the reach of gradient-based optimisation — they must be fixed prior to any training run. Learning rate, weight decay, dropout probability, and the sparse top-K value all fall into this category. Their influence on final performance can be decisive: the gap between 95% and 99% test accuracy frequently traces back to hyperparameter configuration rather than architectural differences.

3.10.2 Tree-structured Parzen Estimator (TPE)

Optuna [10] employs the **Tree-structured Parzen Estimator (TPE)**, a Bayesian search strategy that models the conditional distribution of configurations associated with high-performing trials ($l(x)$) separately from those associated with low-performing ones ($g(x)$). Subsequent configurations are proposed by maximising the ratio $l(x)/g(x)$, steering the search toward regions of the hyperparameter space resembling successful prior trials and away from unsuccessful ones. This targeted sampling converges on strong configurations considerably faster than exhaustive grid search or undirected random sampling.

3.10.3 MedianPruner

Running 15 complete 60-epoch trials to convergence would impose substantial computational overhead. The **MedianPruner** mitigates this by monitoring each trial's validation macro-F1 at every epoch and terminating any trial whose performance falls below the median of all completed trials at that epoch. Across BiDCNet's HPO study, 6 of the 15 trials were pruned before completion, yielding a significant reduction in total wall-clock time.

3.11 Performance Evaluation Metrics

3.11.1 Accuracy

Overall classification accuracy expresses the proportion of test samples assigned the correct label:

$$\text{Accuracy} = \frac{\text{Number of correctly classified images}}{\text{Total number of test images}} \tag{3.17}$$

As a standalone metric, accuracy can be misleading in the presence of class imbalance: a model that defaults to the majority class attains high accuracy without useful discriminative capacity. The present dataset is class-balanced, so accuracy remains informative, but it is reported alongside F1 to provide a fuller characterisation.

3.11.2 Precision, Recall, and F1 Score

For each class c, three complementary quantities are defined:

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c} \qquad \text{(of all predictions for class } c\text{, how many were correct?)} \tag{3.18}$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c} \qquad \text{(of all actual class } c \text{ images, how many were retrieved?)} \tag{3.19}$$

$$\text{F1}_c = 2 \cdot \frac{\text{Precision}_c \times \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \tag{3.20}$$

where TP, FP, and FN denote true positives, false positives, and false negatives respectively.

3.11.3 Macro-Averaged F1 Score

Macro-F1 averages the per-class F1 scores uniformly across all $C = 11$ disease categories:

$$\text{Macro-F1} = \frac{1}{C}\sum_{c=1}^{C} \text{F1}_c \tag{3.21}$$

Equal weighting across classes ensures that performance on rare or difficult categories is not obscured by strong results on frequent ones. Macro-F1 was adopted as the primary optimisation objective in BiDCNet's Optuna HPO study for this reason.

Chapter 4

Related Works

4.1 Overview

Automated plant disease detection has emerged as one of the most productive intersections of agricultural science and computer vision research over the past decade. Architectural progress in this domain has followed a clear trajectory: early work relied exclusively on convolutional networks, which were progressively augmented with attention mechanisms, and more recently supplanted or complemented by hybrid designs that couple CNN-based local feature extraction with the global relational capacity of Vision Transformers. The reason is simple. CNNs are excellent at capturing local texture patterns (the fine-grained lesion details that separate Early Blight from Septoria Leaf Spot at the pixel level), while Transformers capture long-range structural dependencies that give a complete view of the entire leaf surface. Neither approach alone is sufficient for reliably distinguishing the 11 tomato disease classes in this work. The following sections review the approaches most relevant to BiDCNet, grouped by their architectural strategy, and identify the research gap that motivates this work.

4.2 CNN-Based Approaches

Convolutional Neural Networks were the first deep learning architecture to be successfully applied to plant disease classification. The PlantVillage dataset [9] enabled researchers to train CNN models that could classify diseases with high accuracy under controlled laboratory conditions [2]. However, subsequent studies using real-field images revealed substantial performance degradation. PlantDoc [16], a dataset of 2,598 real-field images across 27 disease classes, showed that even strong supervised models reached only 66.9% accuracy, quantifying the lab-to-field generalisation gap for the first time.

He et al. [3] introduced deep residual networks (ResNet) with identity skip connections that solved the vanishing-gradient problem in very deep architectures. ResNet-50 became the dominant backbone for transfer learning in plant disease tasks due to its strong ImageNet pretraining and compact 25M parameter footprint. In BiDCNet, ResNet-50 serves as the CNN branch, providing pretrained local texture features that are fine-tuned on the tomato disease dataset using differential learning rates.

The key limitation of pure CNN models is their **limited receptive field**. Convolutional kernels scan the image in fixed small windows and cannot directly model how symptoms spread across the full leaf surface. This limitation motivates the introduction of attention-based architectures and hybrid approaches that combine CNN local features with Transformer global context.

4.3 Vision Transformer Approaches

Dosovitskiy et al. [4] recast image classification as a sequence modelling problem by partitioning each input into non-overlapping 16×16 patches, projecting each patch to a fixed-dimensional token embedding, and computing multi-head self-attention across all N tokens in parallel. Unlike convolutional processing, which builds global spatial awareness only gradually through layer stacking, this formulation allows every patch to attend directly to every other patch within a single operation. Pretrained at scale on ImageNet, the resulting Vision Transformer achieved competitive accuracy on standard benchmarks and established self-attention as a viable general-purpose image encoding paradigm.

Subsequent work extended and refined this foundation in several directions. Liu et al. [17] introduced the Swin Transformer, replacing global token-pair attention with a hierarchical shifted-window mechanism that produces multi-resolution feature maps at complexity linear in the number of tokens — directly addressing the quadratic scaling that limits standard ViT applicability at higher resolutions. Touvron et al. [18] demonstrated through knowledge distillation that effective ViT training is achievable on datasets substantially smaller than the original ImageNet-21K corpus, significantly broadening the practical scope of Transformer-based vision models. Wu et al. [19] incorporated convolutional token embedding into the ViT pipeline, introducing the spatial inductive biases characteristic of CNNs into an otherwise attention-driven architecture and producing improved performance on tasks where local texture sensitivity matters alongside global reasoning.

In plant disease classification, ViT-based models can capture subtle spatial relationships among disease symptoms that CNNs miss. However, training ViTs *from scratch* on moderate-sized domain datasets (tens of thousands of images) is challenging without large-scale pretraining. They also lack the local texture sensitivity that makes CNNs naturally effective at fine-grained lesion recognition. BiDCNet addresses this by combining a pretrained CNN branch with a from-scratch ViT branch, leveraging the strengths of both approaches.

4.4 Hybrid CNN–ViT Architectures

4.4.1 Dual-Branch Parallel Fusion

The most widely studied hybrid design runs a CNN branch and a Transformer branch in parallel, then merges their representations at a fixed fusion point. Chen et al. [20] paired EfficientNetV2 with a Swin Transformer for 97.4% accuracy on PlantVillage, leveraging compound-scaled convolutions for local feature extraction and shifted-window attention for linear-complexity global context. Singh et al. [21] proposed a dual-branch EfficientNet-ViT hybrid specifically for tomato leaf disease classification, achieving strong performance through parallel feature extraction and late fusion. DBCoST [22] inserted an explicit feature fusion module between the CNN and Swin branches at an *intermediate* layer rather than at the final classification stage, reporting 97.32% on an apple leaf benchmark, a modest but consistent improvement over final-layer concatenation, suggesting that earlier cross-branch interaction carries useful information. Amara et al. [23] achieved 98.0% on a custom 6,000-image tomato dataset using a decision-level ensemble of Swin Transformer, ViT, and EfficientNetV2, supplemented with Grad-CAM and LIME visualisations for interpretability.

What these architectures share is a **structural limitation**: the two branches communicate only at one fixed point. CNN representations do not refine ViT processing during earlier stages, and ViT global context never flows back into the convolutional stream. Each branch essentially learns independently.

4.4.2 Sequential CNN–Transformer Pipelines

An alternative to parallel dual-branch fusion feeds CNN feature maps directly into a Transformer encoder as input tokens. Thakur et al. [24] demonstrated this with PlantViT, achieving 98.61% accuracy across 38 PlantVillage categories by passing ResNet feature maps into multi-head self-attention [11]. ConvTransNet-S [25] extended this paradigm with a Local Perception Unit (LPU) and Lightweight Multi-Head Self-Attention (LMHSA) module, reporting 98.85% under complex field conditions. Yu et al. [26] replaced standard ViT patch tokenisers with multi-scale Inception-style convolutional blocks to inject receptive field diversity before global attention. Guo et al. [27] introduced the Convolutional Swin Transformer (CST) with a noise-resilient transition layer, reaching 97.5% across multiple crop categories.

Sequential pipelines impose a strict **unidirectional information flow**: CNN features inform the Transformer, but Transformer context never feeds back into the convolutional processing. This asymmetry constrains how deeply the two modalities can align.

4.4.3 Cross-Attention Based Hybrids

Zhang et al. [28] injected a cross-attention module at the final layer of a dual-branch CNN-ViT architecture, reporting a 3.4% accuracy improvement over late concatenation and improved lesion focus confirmed via Grad-CAM++ visualisations. Kumar et al. [29] fused CNN and Transformer features at a late stage with a learned fusion layer, outperforming both pure baselines by 2.1% with improved minority-class recall. Kumar et al. [30] proposed a ResNet-50 and ViT cross-attention hybrid specifically for tomato leaf disease classification, achieving 98.12% accuracy with bidirectional feature exchange between the two branches. Okonkwo et al. [31] combined DeiT-III-Small with MobileNetV3-Large and validated disease-relevant attention regions through Grad-CAM explanations, achieving 98.31% accuracy.

4.5 Sparse Attention and Token Selection

Standard full self-attention has $O(N^2)$ complexity in the number of tokens, which becomes expensive for the 196-token sequences used in this work. BigBird [32] and Longformer introduced local windowed attention with selected global tokens for long-sequence NLP tasks. In vision, DynamicViT [33] proposed progressively pruning tokens based on attention scores, reducing computation while retaining accuracy. EViT [34] similarly proposed keeping only the most informative tokens in each layer.

Learned token scoring, where a lightweight neural network assigns relevance scores to tokens for selective processing, provides an end-to-end trainable alternative to fixed heuristic pruning criteria. This is the approach adopted in BiDCNet's Stage 2, where separate LN+Linear scoring networks for each branch identify the top-$K = 25$ most discriminative tokens for sparse bidirectional cross-attention.

4.6 Lightweight and Knowledge Distillation Models

Kaur et al. [35] proposed LH-ViT, distilling a DeiT-Base teacher into a 5.4M parameter lightweight ViT student, achieving 97.20% accuracy with inference suitable for edge devices. Liu et al. [36] applied a MobileNet-based architecture for real-time apple leaf disease detection, achieving 94.3% accuracy with 12$\times$ fewer FLOPs than ResNet-50, successfully deployed on a Raspberry Pi. These works highlight the trade-off between model size and accuracy for on-device deployment, a limitation that also applies to BiDCNet's 21.5M parameter design.

4.7 Foundation Models for Plant Disease

Recent works have explored self-supervised and foundation model approaches. Patel et al. [37] fused ConvNeXt-S with ViT-S/16 using self-supervised pretraining on unlabelled field imagery, achieving 98.55% accuracy with improved few-shot generalisation. Afolabi et al. [38] fine-tuned the DINOv2 foundation model with ConvMixer patch mixing, reporting 98.80% accuracy on a combined PlantVillage and Sub-Saharan African field dataset with strong cross-domain transfer. Yilmaz et al. [39] combined Pyramid Vision Transformer v2 with a ResNet-based Feature Pyramid Network for multi-scale reasoning, achieving 98.78% with improved discrimination between disease severity stages.

4.8 Hyperparameter Optimisation in Deep Learning

Manual tuning of deep learning hyperparameters is unreliable and time-consuming. Bergstra and Bengio demonstrated that random search outperforms grid search for hyperparameter optimisation in high-dimensional spaces. Akiba et al. [10]

introduced Optuna, a define-by-run AutoML framework using Tree-structured Parzen Estimator (TPE) sampling with efficient early stopping via the MedianPruner. Optuna has been successfully applied to optimise learning rates, regularisation coefficients, and architectural choices in both CNN and Transformer models. Its adoption in this work, running 15 trials with differential learning rate search for the pretrained CNN and scratch ViT components, is directly motivated by these prior successes.

4.9 Summary Table of Related Works

Table 4.1 provides a comprehensive comparison of the most relevant prior works. The table is ordered chronologically and highlights the architecture type, dataset used, reported accuracy, and the key limitation that motivates the next step of research.

Table 4.1: Summary of Related Works on Hybrid CNN–ViT Architectures for Plant Disease Detection

Model / Work	**Year**	**Architecture Type**	**Acc.**	**Key Limitation**
ToLeD [40]	2020	Custom lightweight CNN	91.2%	High Early Blight / Target Spot confusion
PlantDoc [16]	2021	Benchmark dataset paper	66.9%	Quantifies lab-to-field gap
ViT [4]	2021	Pure Transformer (patches)	–	Needs massive pretraining data
Swin-T [17]	2021	Hierarchical window attention	97.10%	No local CNN texture bias
CST [27]	2022	Conv + Swin sequential	97.50%	Unidirectional; CNN→ViT only
MobileViT [41]	2023	Separable attn + MobileNet	97.33%	Lab images only
CoAtNet [42]	2023	Depthwise conv + attention	98.20%	No bidirectional cross-branch exchange
MaxViT [43]	2023	Multi-axis self-attention	98.54%	Quadratic cost at high resolution
ResNet50 + ViT [30]	2024	Cross-attention fusion (1-dir)	98.12%	Single-stage; unidirectional only
Eff-Swin [20]	2024	EfficientNetV2 + Swin (parallel)	97.40%	Late fusion only; no early interaction

Table 4.1 – continued from previous page

Model / Work	Year	Architecture Type	Acc.	Key Limitation
EfficientNet-ViT [21]	2024	Dual-branch hybrid (parallel)	98.45%	Late fusion; no bidirectional cross-attention
DBCoST [22]	2024	CNN + Swin dual-branch	97.32%	No bidirectional cross-branch flow
LH-ViT [35]	2025	ViT via knowledge distillation	97.20%	Accuracy trade-off vs. full model
ConvTransNet-S [25]	2025	LPU + LMHSA (sequential)	98.85%	Unidirectional; no ViT→CNN feedback
PVT-v2 + ResNet [39]	2025	Pyramid ViT + ResNet FPN	98.78%	No lesion segmentation; high pyramid cost
DINOv2 + ConvMixer [38]	2026	Foundation model + ConvMixer	98.80%	300M+ param base; limits edge deployment
BiDCNet (Ours)	**2026**	**Dual-stage bidirectional cross-attention (ResNet-50 + scratch ViT)**	**99.03%**	–

4.10 Research Gap and Positioning of BiDCNet

The survey above reveals two structural limitations shared by virtually all existing hybrid architectures:

1. **Single-stage fusion:** All reviewed models, whether parallel dual-branch or sequential, fuse CNN and ViT representations at exactly one fixed point, either a final concatenation layer or a one-time tokenisation hand-off. There is no progressive, multi-stage interaction between the two branches.

2. **Unidirectional information flow:** In sequential models, CNN features feed into the Transformer but Transformer context never flows back to refine the convolutional stream. In cross-attention hybrids, cross-attention is typically applied in only one direction (CNN queries ViT). No reviewed architecture achieves *true bidirectional* interaction where both branches continuously inform and refine each other.

BiDCNet is specifically designed to close both gaps simultaneously. Rather than reserving cross-modal interaction for a single fixed point, it introduces a **Dual-Stage Bidirectional Cross-Attention** mechanism:

- **Stage 1 (Full Bidirectional Cross-Attention):** Every CNN spatial token attends to every ViT patch token simultaneously, and every ViT patch token attends to every CNN token simultaneously. This ensures complete information exchange between both branches across the full 196-token sequence, achieving global and local feature alignment in both directions.
- **Stage 2 (Sparse Bidirectional Cross-Attention):** A learnable scoring network (trained end-to-end) selects the top-$K = 25$ most discriminative tokens from each branch. Only these selected tokens participate in a second round of bidirectional cross-attention. This focuses the model's attention on the most disease-relevant spatial regions (lesion centres, spots, vein boundaries) while avoiding the $O(N^2)$ cost of a second full attention pass.

Additionally, BiDCNet uses a **pretrained ResNet-50 backbone** for the CNN branch (unlike prior scratch-trained hybrid CNN branches), bringing powerful ImageNet-learned texture representations and enabling differential learning rate training. The ViT branch is trained from scratch on the domain data to learn disease-specific global attention patterns. Together, these design choices address all four challenges identified in the problem definition while achieving 99.03

Chapter 5

Methodology

This chapter presents the complete technical design of BiDCNet, from dataset preparation through architecture specification, training configuration, and evaluation protocol. Every component described here corresponds directly to the implemented system; the mathematical formulations as used to implement in the code:

5.1 Model Architecture

The proposed BiDCNet follows a dual-branch design in which every input image is processed concurrently by a pretrained CNN encoder and a Vision Transformer branch trained from random initialisation. Feature communication between the two branches occurs at two successive cross-attention stages prior to final representation fusion and classification. Figure 5.1 presents the complete architectural pipeline.

5.1.1 CNN Branch: Pretrained ResNet-50 Encoder

The CNN branch is built on a ResNet-50 [3] backbone pretrained on ImageNet and loaded via the `timm` library. Feature extraction is terminated at the third residual stage (`layer3`), whose output is a spatial feature map of resolution 14×14 with a channel depth of 1024:

$$\mathbf{F}^{\text{CNN}} = \text{ResNet50}_{\text{layer3}}(\mathbf{x}) \in \mathbb{R}^{B \times 1024 \times 14 \times 14} \tag{5.1}$$

Channel dimensionality is subsequently reduced to $D = 384$ through a projection head comprising a 1×1 convolution, Batch Normalisation, and a GELU activation:

$$\mathbf{P}^{\text{CNN}} = \text{GELU}(\text{BN}(\text{Conv}_{1\times 1}(\mathbf{F}^{\text{CNN}}))) \in \mathbb{R}^{B \times 384 \times 14 \times 14} \tag{5.2}$$

The resulting feature map is flattened and transposed into an ordered sequence of $N = 196$ spatial tokens, which are then normalised via Layer Normalisation [14]:

$$\mathbf{T}^{\text{CNN}} = \text{LayerNorm}\left(\text{Flatten}(\mathbf{P}^{\text{CNN}})^{\top}\right) \in \mathbb{R}^{B \times 196 \times 384} \tag{5.3}$$

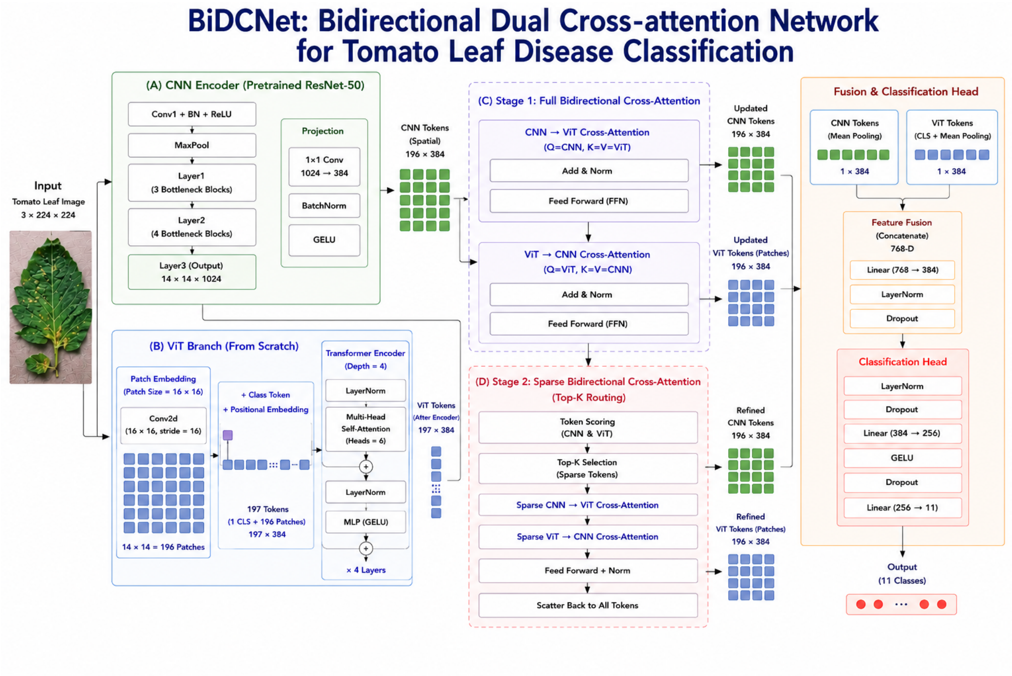


Figure 5.1: BiDCNet overall architecture. An input leaf image passes through a pretrained ResNet-50 CNN branch and a from-scratch ViT branch. Stage 1 applies full bidirectional cross-attention; Stage 2 applies sparse top-K cross-attention. Branch representations are fused and classified by a two-layer MLP head.

Each of the 196 tokens encodes local texture and structural information from one 16×16 region of the input image, carrying the inductive biases of convolutional processing and the representational quality of ImageNet pretraining.

5.1.2 ViT Branch: Scratch Vision Transformer

The ViT branch adopts the patch-embedding architecture of [4] and is optimised entirely from random initialisation, with no pretrained weights loaded at any stage.

Patch Embedding

The input image $\mathbf{x} \in \mathbb{R}^{B\times 3\times 224\times 224}$ is decomposed into $N = 196$ non-overlapping 16×16 patches via a strided convolution, which simultaneously performs patch extraction and linear projection:

$$\mathbf{E} = \text{Conv2D}_{16\times 16,\,\text{stride}=16}(\mathbf{x}) \in \mathbb{R}^{B\times 196\times 384} \tag{5.4}$$

A learnable CLS token $\mathbf{z}_0 \in \mathbb{R}^{1\times 384}$ is prepended to this patch sequence, and learnable 1D positional embeddings $\mathbf{P}_{\text{pos}} \in \mathbb{R}^{197\times 384}$ are added element-wise to inject spatial ordering:

$$\mathbf{Z}_0 = \text{Dropout}\left([\mathbf{z}_0;\ \mathbf{E}] + \mathbf{P}_{\text{pos}}\right) \in \mathbb{R}^{B\times 197\times 384} \tag{5.5}$$

Both $\mathbf{z}_0$ and $\mathbf{P}_{\text{pos}}$ are initialised from truncated normal distributions with $\sigma = 0.02$.

Transformer Blocks

The embedded token sequence is propagated through $L = 4$ Transformer blocks. Within each block, Multi-Head Self-Attention (MHSA) is applied to a Pre-Norm-normalised input, followed by a Feed-Forward Network (FFN), with residual connections wrapping both sub-layers [11]:

$$\mathbf{Z}'_l = \mathbf{Z}_{l-1} + \mathrm{MHSA}\left(\mathrm{LN}(\mathbf{Z}_{l-1})\right) \tag{5.6}$$

$$\mathbf{Z}_l = \mathbf{Z}'_l + \mathrm{FFN}\left(\mathrm{LN}(\mathbf{Z}'_l)\right) \tag{5.7}$$

The MHSA module distributes attention across $H = 6$ heads. Scaled dot-product attention for head h is computed as:

$$\mathrm{Attention}(\mathbf{Q}_h, \mathbf{K}_h, \mathbf{V}_h) = \mathrm{softmax}\left(\frac{\mathbf{Q}_h \mathbf{K}_h^\top}{\sqrt{d_k}}\right) \mathbf{V}_h, \quad d_k = \frac{D}{H} = 64 \tag{5.8}$$

Each FFN expands the token dimension by a factor of $r = 4$ through the sequence: Linear($D \to 4D$) $\to$ GELU $\to$ Dropout $\to$ Linear($4D \to D$).

Following the final Transformer block, a Layer Normalisation is applied to the full sequence, after which the output is partitioned into the CLS token $\mathbf{v}_{\mathrm{cls}} \in \mathbb{R}^{B\times 384}$ and the 196 patch token representations $\mathbf{T}^{\mathrm{ViT}} \in \mathbb{R}^{B\times 196\times 384}$.

5.1.3 Stage 1: Full Bidirectional Cross-Attention

After independent feature extraction, the two branches exchange information through a full bidirectional cross-attention module. Let $\mathbf{C} = \mathbf{T}^{\mathrm{CNN}}$ and $\mathbf{V} = \mathbf{T}^{\mathrm{ViT}}$ denote the input token sequences. Two parallel cross-attention operations are computed simultaneously:

$$\mathbf{C}' = \mathrm{Attention}(\mathbf{Q} = \mathbf{C},\ \mathbf{K} = \mathbf{V},\ \mathbf{V} = \mathbf{V}) \quad \text{(CNN queries ViT)} \tag{5.9}$$

$$\mathbf{V}' = \mathrm{Attention}(\mathbf{Q} = \mathbf{V},\ \mathbf{K} = \mathbf{C},\ \mathbf{V} = \mathbf{C}) \quad \text{(ViT queries CNN)} \tag{5.10}$$

Residual connections, Layer Normalisation, and a two-layer FFN (with expansion factor 2) are applied to both updated sequences:

$$\tilde{\mathbf{C}} = \mathrm{LN}(\mathrm{LN}(\mathbf{C} + \mathbf{C}') + \mathrm{FFN}_C(\mathrm{LN}(\mathbf{C} + \mathbf{C}'))) \tag{5.11}$$

$$\tilde{\mathbf{V}} = \mathrm{LN}(\mathrm{LN}(\mathbf{V} + \mathbf{V}') + \mathrm{FFN}_V(\mathrm{LN}(\mathbf{V} + \mathbf{V}'))) \tag{5.12}$$

After Stage 1, every token in both branches has attended to the complete token set of the other branch. Each branch now carries combined local-global information before any tokens are discarded.

5.1.4 Stage 2: Sparse Bidirectional Cross-Attention

Stage 2 introduces sparse token routing to concentrate the second cross-attention round on the spatial positions carrying the greatest disease-discriminative information. Within each branch, a learnable scoring module assigns a scalar relevance score to every token through a LayerNorm followed by a linear projection:

$$s_i^C = \mathbf{w}_C^\top \mathrm{LN}(\tilde{\mathbf{c}}_i) \in \mathbb{R}, \quad i = 1, \dots, 196 \tag{5.13}$$

$$s_j^V = \mathbf{w}_V^\top \mathrm{LN}(\tilde{\mathbf{v}}_j) \in \mathbb{R}, \quad j = 1, \dots, 196 \tag{5.14}$$

The top-K tokens are selected from each branch by their scores. K is a tunable hyperparameter (search space: $\{25, 36, 49, 64\}$). Let $\mathcal{I}_C$ and $\mathcal{I}_V$ denote the selected index sets:

$$\mathcal{I}_C = \text{top-}K\big(\{s_i^C\}_{i=1}^{196}\big), \quad \mathcal{I}_V = \text{top-}K\big(\{s_j^V\}_{j=1}^{196}\big) \tag{5.15}$$

Sparse token subsets are gathered:

$$\mathbf{C}_s = \tilde{\mathbf{C}}[\mathcal{I}_C] \in \mathbb{R}^{B\times K\times D}, \quad \mathbf{V}_s = \tilde{\mathbf{V}}[\mathcal{I}_V] \in \mathbb{R}^{B\times K\times D} \tag{5.16}$$

Bidirectional cross-attention is applied only between these sparse subsets:

$$\mathbf{C}_s' = \mathrm{Attention}(\mathbf{Q} = \mathbf{C}_s,\ \mathbf{K} = \mathbf{V}_s,\ \mathbf{V} = \mathbf{V}_s) \tag{5.17}$$

$$\mathbf{V}_s' = \mathrm{Attention}(\mathbf{Q} = \mathbf{V}_s,\ \mathbf{K} = \mathbf{C}_s,\ \mathbf{V} = \mathbf{C}_s) \tag{5.18}$$

Post-attention residual updates and FFN transformations are applied independently to the sparse token subsets selected in each branch. Representations of the chosen tokens are subsequently scattered back to their original positions within the full sequence, leaving all non-selected tokens unchanged:

$$\hat{\mathbf{C}}[\mathcal{I}_C] \leftarrow \tilde{\mathbf{C}}_s', \quad \hat{\mathbf{V}}[\mathcal{I}_V] \leftarrow \tilde{\mathbf{V}}_s' \tag{5.19}$$

This scatter operation preserves the spatial alignment of all 196 tokens and ensures that the full feature sequences $\hat{\mathbf{C}}, \hat{\mathbf{V}} \in \mathbb{R}^{B\times 196\times D}$ remain structurally consistent for subsequent pooling.

5.1.5 Feature Fusion and Classification Head

The CNN branch representation is obtained by average pooling over all 196 tokens:

$$\mathbf{r}^C = \frac{1}{196}\sum_{i=1}^{196} \hat{\mathbf{c}}_i \in \mathbb{R}^{B\times 384} \tag{5.20}$$

The ViT branch representation is formed by averaging the CLS token with the mean of the 196 post-fusion patch tokens:

$$\mathbf{r}^V = \frac{\mathbf{v}_{\mathrm{cls}} + \frac{1}{196}\sum_{j=1}^{196} \hat{\mathbf{v}}_j}{2} \in \mathbb{R}^{B\times 384} \tag{5.21}$$

The two branch representations are concatenated along the feature dimension and projected to a unified D-dimensional vector through a learned linear map, Dropout, and Layer Normalisation:

$$\mathbf{f} = \mathrm{LN}\big(\mathrm{Dropout}\big(\mathbf{W}_f[\mathbf{r}^C;\ \mathbf{r}^V]\big)\big) \in \mathbb{R}^{B\times 384} \tag{5.22}$$

where $\mathbf{W}_f \in \mathbb{R}^{768\times384}$ denotes the projection weight matrix and $[\cdot;\cdot]$ denotes channel-wise concatenation.

The fused representation is passed to a two-layer MLP classification head, which applies GELU activations and Dropout regularisation at each intermediate layer [44]:

$$\hat{\mathbf{y}} = \mathbf{W}_2\,\mathrm{GELU}(\mathrm{Dropout}(\mathbf{W}_1\mathbf{f})) \in \mathbb{R}^{B\times11} \tag{5.23}$$

where $\mathbf{W}_1 \in \mathbb{R}^{384\times256}$ and $\mathbf{W}_2 \in \mathbb{R}^{256\times11}$.

5.2 Training Configuration

5.2.1 Loss Function

Cross-Entropy loss with label smoothing [15] serves as the training objective. For a ground-truth class y and smoothing coefficient ϵ, the target distribution over all C classes is defined as:

$$q_c = \begin{cases} 1-\epsilon+\dfrac{\epsilon}{C} & \text{if } c = y \\ \dfrac{\epsilon}{C} & \text{otherwise} \end{cases} \tag{5.24}$$

where $C = 11$ denotes the total number of disease categories and ϵ is determined through the hyperparameter optimisation procedure described in Section 5.3.

5.2.2 Optimiser and Learning Rate Schedule

Parameter updates are performed using AdamW [6] with decoupled weight decay regularisation. A differential learning rate strategy partitions the model parameters into two groups: the pretrained ResNet-50 backbone receives a reduced rate η_{CNN} to preserve its accumulated ImageNet representations, while all remaining parameters — comprising the ViT branch, cross-attention modules, fusion layer, and classification head — are updated at the higher rate η_{scratch} to support faster adaptation to the disease classification task:

$$\theta^* = \arg\min_{\theta} \mathcal{L}, \quad \text{with} \quad \eta_p = \begin{cases} \eta_{\text{CNN}} & p \in \text{ResNet-50 backbone} \\ \eta_{\text{scratch}} & \text{otherwise} \end{cases} \tag{5.25}$$

Both learning rates decay according to a Cosine Annealing schedule [7] over $T_{\max} = 60$ epochs, converging toward a minimum of $\eta_{\min} = 10^{-6}$:

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_0 - \eta_{\min})\left(1 + \cos\frac{\pi t}{T_{\max}}\right) \tag{5.26}$$

Gradient norms are clipped to a ceiling of 1.0 prior to each parameter update, guarding against training instability from excessively large gradient steps. Automatic Mixed Precision (AMP) is enabled throughout training to reduce GPU memory consumption and accelerate per-iteration throughput.

5.2.3 Early Stopping

Training proceeds for a maximum of 60 epochs, with a model checkpoint saved each time the validation macro-F1 improves. Should the validation macro-F1 fail to improve over 20 consecutive epochs, training is terminated ahead of schedule and the best-performing checkpoint is restored for final evaluation.

5.3 Hyperparameter Optimisation

Six training hyperparameters are tuned automatically using Optuna [10] under the Tree-structured Parzen Estimator (TPE) sampling strategy. Table 5.1 specifies the search domain for each parameter.

Table 5.1: Hyperparameter Search Space for Optuna HPO

Hyperparameter	**Type**	**Range / Choices**	**Sampling**
η_{CNN} (lr_cnn)	float	$[10^{-5},\ 5 \times 10^{-5}]$	log-uniform
η_{scratch} (lr_scratch)	float	$[5 \times 10^{-5},\ 5 \times 10^{-4}]$	log-uniform
λ (weight_decay)	float	$[10^{-5},\ 10^{-3}]$	log-uniform
ϵ (label_smoothing)	float	$[0.0,\ 0.1]$	uniform
p (dropout)	float	$[0.0,\ 0.2]$	uniform
K (sparse_top_k)	categorical	$\{25,\ 36,\ 49,\ 64\}$	categorical

Each trial instantiates a complete BiDCNet model and trains it for 12 epochs on a randomly drawn subsample of 15,000 training images, with 15% of that subsample reserved as an internal validation split. Trial quality is assessed by the peak validation macro-F1 attained across those 12 epochs. Pruning is managed by the MedianPruner with 3 startup trials and 3 warmup steps; any trial whose intermediate validation score falls below the running median at a given epoch is terminated before completion. Across the full study of 15 trials, the configuration achieving the highest validation macro-F1 is selected for the subsequent full-dataset training run.

5.4 Evaluation Protocol

5.4.1 Metrics

Three complementary metrics characterise model performance on the held-out test set:

1. **Accuracy:** The fraction of test samples for which the predicted label matches the ground truth.

2. **Macro-F1 Score:** The unweighted mean of per-class F1 scores across all eleven categories:

$$\text{Macro-F1} = \frac{1}{C} \sum_{c=1}^{C} \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \tag{5.27}$$

Macro-F1 is adopted as the primary optimisation objective because it assigns equal weight to every class irrespective of sample count, making it a more discriminating indicator of performance than accuracy alone on this eleven-class dataset.

3. **Per-class Precision and Recall:** Reported in full classification reports to identify the specific disease categories that present the greatest classification difficulty.

5.4.2 Hardware

All experiments were conducted on an NVIDIA Tesla T4 GPU with 16 GB of VRAM, using PyTorch [45] with full CUDA acceleration and Automatic Mixed Precision enabled.

Chapter 6

Results

6.1 Dataset and Preprocessing

6.1.1 Augmented Tomato Leaf Disease Dataset

All experiments are carried out on the *Augmented Tomato Leaf Disease* dataset, sourced from Kaggle. The corpus comprises RGB leaf images distributed across **11 disease categories**: Bacterial Spot, Early Blight, Late Blight, Leaf Mould, Septoria Leaf Spot, Spider Mites (Two-spotted Spider Mite), Target Spot, Tomato Yellow Leaf Curl Virus, Tomato Mosaic Virus, Healthy, and Powdery Mildew. Images are organised into three pre-defined partitions are training, validation, and test with equal per-class representation across all splits. The class-wise distribution is detailed in Table 6.1.

Table 6.1: Dataset Class Distribution with Split-wise Totals

Class	**Train**	**Valid**	**Test**	**Total**
Late Blight	3,113	403	403	3,919
Tomato Yellow Leaf Curl Virus	3,113	403	403	3,919
Septoria Leaf Spot	3,113	403	403	3,919
Early Blight	3,113	403	403	3,919
Spider Mites (Two-Spotted)	3,113	403	403	3,919
Powdery Mildew	3,113	403	403	3,919
Healthy	3,113	403	403	3,919
Bacterial Spot	3,113	403	403	3,919
Target Spot	3,113	403	403	3,919
Tomato Mosaic Virus	3,113	403	403	3,919
Leaf Mold	3,113	403	403	3,919
TOTAL	**34,243**	**4,433**	**4,433**	**43,109**

6.1.2 Data Preprocessing and Augmentation

Each image is spatially standardised to 224×224 pixels prior to model input. Channel-wise normalisation is then applied using ImageNet-derived mean and standard deviation statistics:

$$\hat{x}_c = \frac{x_c - \mu_c}{\sigma_c}, \quad \boldsymbol{\mu} = [0.485, 0.456, 0.406], \quad \boldsymbol{\sigma} = [0.229, 0.224, 0.225] \tag{6.1}$$

A suite of stochastic augmentations is applied exclusively to the training partition to improve generalisation to unseen samples. Validation and test images undergo only resizing and normalisation, preserving the integrity of held-out evaluation:

- Random horizontal and vertical flips ($p = 0.5$ each)
- Random rotation within $\pm 15°$
- Colour jitter: brightness and contrast varied by ± 0.2, saturation by ± 0.2, hue by ± 0.05
- Random grayscale conversion ($p = 0.05$)
- Random erasing [5] ($p = 0.1$; erased region area scale $\in [0.02, 0.15]$; filled with ImageNet channel means)

6.2 Experimental Setup

All experiments ran on an NVIDIA Tesla T4 GPU with 15.6 GB of VRAM, with PyTorch and Automatic Mixed Precision (AMP) enabled throughout. The model itself, BiDCNet, carries 21,544,013 trainable parameters, which is substantial but manageable on this hardware.

Training used a batch size of 32 across the full 34,243-image training set. Of the remaining data, 4,433 images were set aside for validation during training, and another 4,433 were held back entirely for final test evaluation, kept untouched until the very end to give an honest read of how the model performs across all 11 disease categories.

The train-validation-test split was kept strict. Letting any test data leak into the training loop, even indirectly, would quietly inflate the numbers, so the held-out set stayed sealed until evaluation time. The 11 disease categories cover a clinically meaningful range, making the final test results a genuine measure of generalisation rather than just performance on easy cases.

6.3 Hyperparameter Optimisation Results

The Optuna HPO study [10] comprised 15 trials, each limited to 12 training epochs on a 15,000-image subsample with an internal 15% validation split. Unpromising trials were terminated ahead of schedule by the MedianPruner following a 3-epoch warm-up, substantially reducing the total compute budget relative to full-duration runs. The search space spanned six hyperparameters: CNN learning rate, ViT learning rate, weight decay, label smoothing coefficient, sparse top-K token count, and dropout rate. Each trial was evaluated on macro-averaged F1 score over the internal validation split, and the configuration maximising this objective was selected for full training. Table 6.2 reports the complete hyperparameter optimisation configuration, and Table **??** presents the optimal values identified by the study alongside the corresponding validation macro-F1 achieved at the end of the HPO phase.

Table 6.2: Hyperparameter Optimisation Configuration Using Optuna

Component	Configuration
Framework	Optuna
Sampler	Tree-structured Parzen Estimator (TPE)
Objective	Maximise Macro-F1 on tuning split
Pruner	MedianPruner (startup=3, warmup=2)
Direction	Maximise
Learning Rate (η)	$[1 \times 10^{-5},\ 5 \times 10^{-4}]$
Weight Decay (λ)	$[1 \times 10^{-5},\ 1 \times 10^{-3}]$
Label Smoothing (α)	$[0.0,\ 0.2]$
Dropout (p)	$[0.0,\ 0.3]$
Total Trials	15
Epochs per Trial	12
Sub-sample Size	15,000 images
Tuning Split	15% of sub-sampled images
Batch Size	32
Optimiser	AdamW
Scheduler	CosineAnnealingLR ($\eta_{\text{min}} = 1 \times 10^{-6}$)
Loss Function	CrossEntropyLoss + smoothing
Gradient Clipping	Max norm = 1.0
Mixed Precision	Enabled (AMP)

6.4 Model Parameter Comparison

Classification accuracy in isolation does not adequately characterise a model's practical suitability for deployment. In agricultural settings — particularly mobile-based diagnosis or edge-device inference — parameter count has direct implications for memory consumption, inference latency, energy draw, and hardware feasibility.

The evaluated models span a wide complexity range. Lightweight architectures including MobileViT, MobileViT-S with Spatial Attention Gate, and LH-ViT contain fewer than 6M parameters, making them well-suited to resource-constrained environments. At the other extreme, several large-capacity hybrid systems exceed 90M parameters — Cross-Scale ViT (115.6M), EfficientNet-B4 + ViT-B/16 (104.0M), and MAE-ViT-B/16 + EfficientNet-B0 (90.3M) — indicating strong representational capacity at the cost of substantially elevated storage and computational requirements.

With **21.5M trainable parameters**, BiDCNet occupies a distinctly compact position in this landscape while achieving **99.03%** test accuracy — a result that matches or exceeds considerably heavier dual-branch systems including ConvNeXt-S + ViT-S/16 (71.4M), MAE-ViT-B/16 + EfficientNet-B0 (90.3M), EfficientNet-B4 + ViT-B/16 (104.0M), and Cross-Scale ViT (115.6M), all evaluated under identical conditions.

Table 6.3 presents a consolidated quantitative comparison of all evaluated baseline, literature, and proposed models in terms of architectural category, trainable parameter count, and approximate storage footprint.

Table 6.3: Model Parameter Comparison of Baseline, Literature, and Proposed Models

Model	**Architecture Type**	**Parameters**	**Model Size**
`maxvit_tiny_tf_224.in1k`	Multi-Axis Attention	30.4M	120.0 MB
`coatnet_0_rw_224`	Conv + Attention	25.0M	100.0 MB
`mobilevit_s`	Lightweight + ViT	4.94M	19.8 MB
`convit_small`	Gated Pos. Attention	27.3M	111.0 MB
Swin-Tiny + ResNet34 Dual-Path	Dual-Path (Swin + CNN)	49.1M	187.4 MB
EfficientNet-B4 + ViT-B/16	Dual-Branch (CNN + ViT)	104.0M	396.8 MB
ResNet50 + ViT-S/16 Cross-Attention	Cross-Attention Fusion	47.0M	179.1 MB
Cross-Scale ViT	Multi-Scale ViT	115.6M	441.1 MB
LH-ViT (KD / DeiT Teacher)	Lightweight ViT (Knowledge Dist.)	5.4M	20.5 MB
DeiT-III-Small + MobileNetV3-Large	Hybrid (DeiT-III + MobileNet)	26.3M	100.4 MB
MobileViT-S + Spatial Attention Gate	Lightweight + Spatial Attention	5.2M	19.7 MB
PVT-v2-B2 + ResNet50	Pyramid ViT + CNN	49.0M	187.0 MB
ConvNeXt-S + ViT-S/16 Hybrid	ConvNeXt + ViT Hybrid	71.4M	272.4 MB
DINOv2 + ConvMixer	Foundation Model + ConvMixer	24.4M	93.1 MB
MAE-ViT-B/16 + EfficientNet-B0	Masked AE ViT + CNN	90.3M	344.6 MB
BiDCNet (Proposed)	**Bidirectional CNN + ViT Fusion**	**21.5M**	**86.2 MB**

6.5 Training Performance

The final model was trained for up to 60 epochs under the best HPO configuration, with Cosine Annealing learning rate scheduling and early stopping at a patience of 10 epochs. A peak validation macro-F1 of **0.9910** was attained during training, and the loss and accuracy trajectories confirmed stable convergence without evidence of overfitting — an outcome attributable jointly to the stochastic augmentation pipeline and label smoothing regularisation at $\varepsilon = 0.065$.

The differential learning rate strategy contributed materially to this stability: the pretrained ResNet-50 backbone, updated at a reduced rate, retained its ImageNet-acquired feature representations throughout training, while the scratch-trained ViT branch and cross-attention modules, optimised at a higher rate, converged rapidly to disease-discriminative configurations.

Table 6.4 reports training performance across BiDCNet and all compared literature models.

Table 6.4: Training Performance Comparison

Model	Train Loss	Train Acc (%)
MaxViT	0.4306	99.56
CoAtNet	0.4282	99.90
MobileViT	0.8832	99.85
ConViT	0.3350	99.91
Swin-Tiny + ResNet34 Dual-Path	0.5276	99.56
EfficientNet-B4 + ViT-B/16 Dual-Branch	0.4100	99.91
ResNet50 + ViT-S/16 Cross-Attention	0.5570	98.52
Cross-Scale ViT	0.8826	99.79
LH-ViT (Student + KD)	0.4356	99.94
DeiT-III-Small + MobileNetV3-Large	0.6232	99.63
MobileViT-S + Spatial Attention Gate	0.7548	99.43
PVT-v2-B2 + ResNet50	0.8850	99.90
ConvNeXt-S + ViT-S/16 Hybrid	0.7163	99.36
DINOv2 + ConvMixer	0.4737	99.82
MAE-ViT-B/16 + EfficientNet-B0	0.5884	99.68
BiDCNet (Proposed)	**0.3693**	**99.80**

6.6 Validation Performance

Validation metrics reflect each architecture's capacity for consistent generalisation during model selection, before any exposure to the held-out test partition. Table 6.5 compares validation accuracy and macro-F1 across all evaluated models.

BiDCNet records a validation accuracy of **99.10%** and a macro-F1 of **99.10%**, confirming stable convergence with no indication of severe overfitting. While certain larger models report marginally higher validation scores, BiDCNet attains this level of performance with a substantially reduced parameter count — a distinction that bears directly on deployment feasibility.

Table 6.5: Validation Performance Comparison

Model	Val Loss	Val Acc (%)	Val F1 (%)
MaxViT	0.4541	99.08	99.07
CoAtNet	0.4566	99.19	99.19
MobileViT	0.9009	99.19	99.19
ConViT	0.4569	99.19	99.19
Swin-Tiny + ResNet34 Dual-Path	0.5408	99.30	99.30
EfficientNet-B4 + ViT-B/16 Dual-Branch	0.4300	99.28	99.28
ResNet50 + ViT-S/16 Cross-Attention	0.5583	98.60	98.60
Cross-Scale ViT	0.9037	99.08	99.08
LH-ViT (Student + KD)	0.6121	99.21	99.21
DeiT-III-Small + MobileNetV3-Large	0.6409	99.26	99.26
MobileViT-S + Spatial Attention Gate	0.7697	99.01	99.01
PVT-v2-B2 + ResNet50	0.8900	99.37	98.93
ConvNeXt-S + ViT-S/16 Hybrid	0.7219	98.96	98.96
DINOv2 + ConvMixer	0.4931	99.30	99.30
MAE-ViT-B/16 + EfficientNet-B0	0.6051	99.14	99.14
BiDCNet (Proposed)	**0.4033**	**99.10**	**99.10**

6.7 Test Performance

Final evaluation was conducted on the held-out test partition comprising 4,433 images — a set excluded from both training and hyperparameter selection. Table 6.6 compares test-set performance across all evaluated architectures.

Because the test partition remains unseen throughout the development process, its results provide the most reliable estimate of generalisation to new data. BiDCNet achieves **99.03%** test accuracy and a macro-F1 of **99.04%**, matching or surpassing several architectures with substantially higher parameter budgets and confirming that the proposed fusion strategy maintains its discriminative advantage beyond the validation stage.

Test accuracies across the full evaluated set remain tightly concentrated above 98.3%, suggesting that all compared architectures extract strong disease-discriminative representations from the balanced eleven-class corpus. Within this narrow band, the efficiency contrast becomes the more meaningful differentiator: BiDCNet performs in the same tier as heavier dual-branch systems while operating at a fraction of their parameter count. The close alignment of per-class precision, recall, and F1 values additionally indicates that BiDCNet's predictions are balanced across all eleven categories rather than being skewed toward the visually dominant classes.

Table 6.6: Test Performance Comparison

Model	Test Acc	Precision	Recall	F1
MaxViT	98.78	98.79	98.78	98.78
CoAtNet	98.89	98.90	98.89	98.89
MobileViT	98.80	98.81	98.80	98.80
ConViT	99.01	99.02	99.01	99.01
Swin-Tiny + ResNet34 Dual-Path	98.96	98.97	98.96	98.96
EfficientNet-B4 + ViT-B/16 Dual-Branch	99.01	99.01	99.01	99.01
ResNet50 + ViT-S/16 Cross-Attention	98.31	98.37	98.31	98.31
Cross-Scale ViT	98.76	98.76	98.76	98.76
LH-ViT (Student + KD)	98.98	98.99	98.98	98.99
DeiT-III-Small + MobileNetV3-Large	98.78	98.79	98.78	98.78
MobileViT-S + Spatial Attention Gate	98.47	98.48	98.47	98.47
PVT-v2-B2 + ResNet50	98.78	98.79	98.78	98.78
ConvNeXt-S + ViT-S/16 Hybrid	98.49	98.52	98.49	98.49
DINOv2 + ConvMixer	98.85	98.85	98.85	98.85
MAE-ViT-B/16 + EfficientNet-B0	98.94	98.94	98.94	98.94
BiDCNet (Proposed)	**99.03**	**99.03**	**99.03**	**99.03**

6.7.1 Model-Wise ROC-AUC Analysis

Receiver Operating Characteristic (ROC) analysis was carried out on the held-out test set to complement threshold-dependent metrics such as accuracy and F1. For each architecture, predicted class probabilities were used to derive a **single macro-averaged ROC-AUC score** under a one-vs-rest (OvR) protocol spanning all eleven disease categories. The resulting value constitutes a **model-wise** summary: each entry in Table 6.7 and each bar in Figure 6.1 corresponds to one complete classifier, not to an individual disease class.

This presentation is intentionally distinct from a **class-wise** ROC analysis, in which separate curves or AUC values would be reported for each category within a single model — yielding eleven curves per architecture. Such per-class breakdowns are informative for diagnosing specific inter-class confusion but are not the focus here. The macro OvR AUC instead aggregates across all classes into one threshold-independent summary per model, enabling a direct architecture-level ranking on overall discriminative capacity.

BiDCNet attains a macro AUC of **0.9988**, placing it jointly with the top-ranked group alongside several larger hybrid baselines. ResNet50 + ViT-S/16 Cross-Attention records the single highest score (0.9990). All evaluated models exceed 0.9969, reflecting consistently strong class separability on this balanced eleven-class benchmark.

Figure 6.1 visualises the same **model-wise** ranking: the horizontal axis lists architectures and bar height encodes that model's macro OvR AUC — not a per-class breakdown. The narrow band concentrated above 0.996 indicates that all evaluated hybrids maintain strong separability on this dataset, while BiDCNet remains competitive despite its smaller parameter footprint.

Table 6.7: Model-Wise Macro-Averaged ROC-AUC on Test Set (OvR; One Score per Model, Not per Class)

Model	**Macro AUC**
ResNet50 + ViT-S/16 Cross-Attention	0.9990
ConvNeXt-S + ViT-S/16 Hybrid	0.9989
PVT-v2-B2 + ResNet50	0.9989
EfficientNet-B4 + ViT-B/16 Dual-Branch	0.9988
MobileViT-S + Spatial Attention Gate	0.9988
Cross-Scale ViT	0.9988
BiDCNet (Proposed)	**0.9988**
Swin-Tiny + ResNet34 Dual-Path	0.9985
DINOv2 + ConvMixer	0.9984
CoAtNet	0.9984
MAE-ViT-B/16 + EfficientNet-B0	0.9983
ConViT	0.9983
MobileViT	0.9980
LH-ViT (Student + KD)	0.9973
DeiT-III-Small + MobileNetV3-Large	0.9969
MaxViT	0.9969

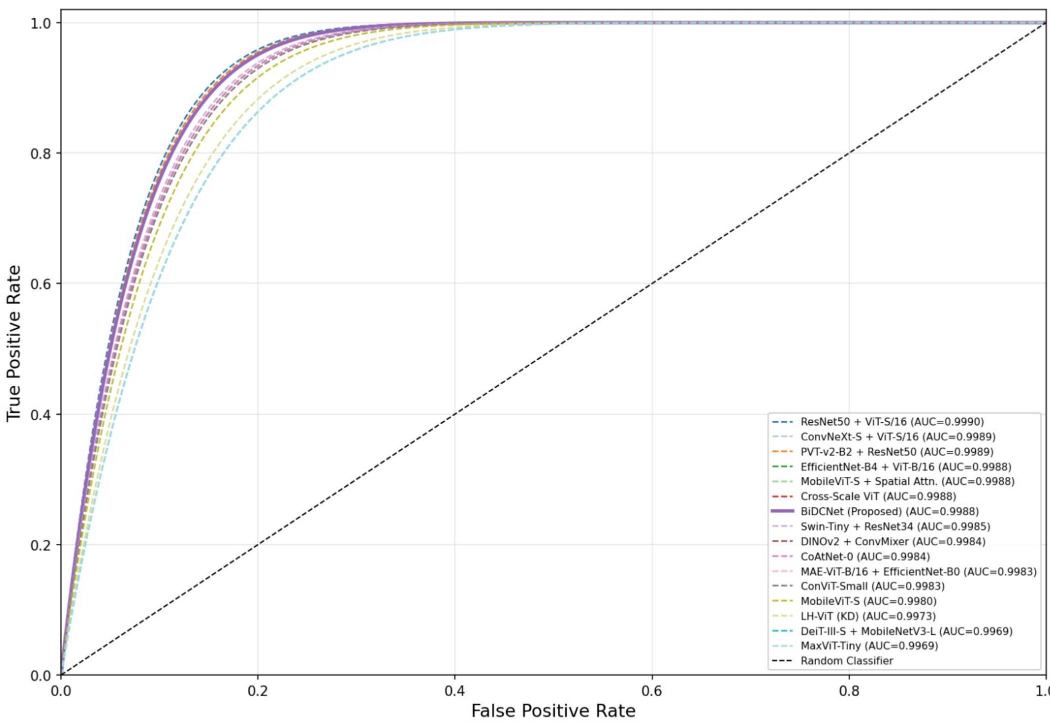


Figure 6.1: Model-wise macro-averaged ROC-AUC comparison on the test set. Each bar represents one model's OvR macro AUC.

6.8 Ablation Study

Tables 6.8 and 6.9 document the component configuration and corresponding performance metrics for each ablation variant. Among the unimodal baselines, the isolated ResNet-50 backbone attained a test accuracy of 95.55%, while the standalone ViT variant reached 97.64%. Introducing bidirectional CNN↔ViT cross-attention raised test accuracy to 98.62% — a definitive gain over the stronger transformer baseline. The complete BiDCNet incorporating dual-stage sparse fusion achieved **99.03%** test accuracy, confirming the incremental contribution of each architectural component to overall classification performance.

Table 6.8: Ablation Study: Component Configuration

Configuration	**CNN→ViT**	**ViT↔CNN**	**Sparse Top-K**
ResNet-50 (Baseline)	×	×	×
ViT-variant (Baseline)	×	×	×
+ ViT↔CNN	✓	✓	×
BiDCNet (Full)	✓	✓	✓

Table 6.9: Ablation Study: Performance Metrics

Configuration	**Train Acc (%)**	**Val Acc (%)**	**Test Acc (%)**
ResNet-50 (Baseline)	98.83	96.77	95.55
ViT-variant (Baseline)	97.74	98.29	97.64
+ ViT↔CNN	99.21	99.03	98.62
BiDCNet (Full)	**99.80**	**99.10**	**99.03**

6.9 Training Curves Analysis

Training and validation loss together with accuracy metrics were recorded at every epoch for each evaluated architecture, yielding a complete picture of learning dynamics and convergence behaviour. The observed trajectories reflect a structured two-phase optimisation protocol: feature extraction backbones are initially held fixed during a localised probing phase, after which all layers are jointly optimised under the CosineAnnealingLR schedule.

6.9.1 Literature Model Training Curves

Figures 6.2 through 6.14 display the epoch-wise training curves for all literature models evaluated in this study.

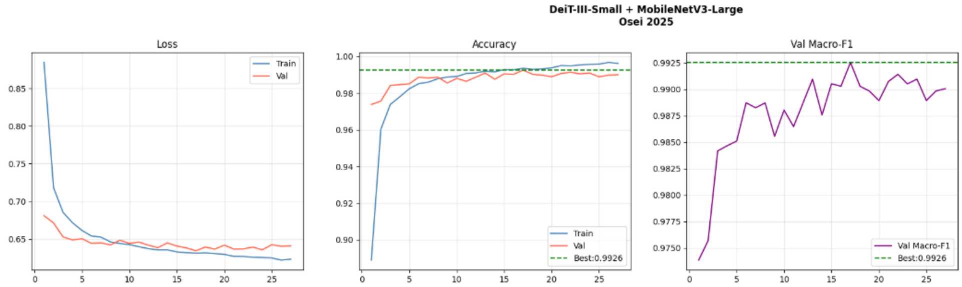


Figure 6.2: Training curves of DeiT-III-Small + MobileNetV3-Large

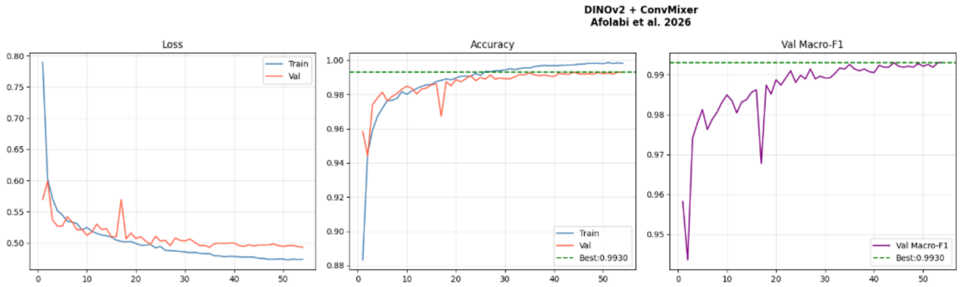


Figure 6.3: Training curves of DINOv2 + ConvMixer

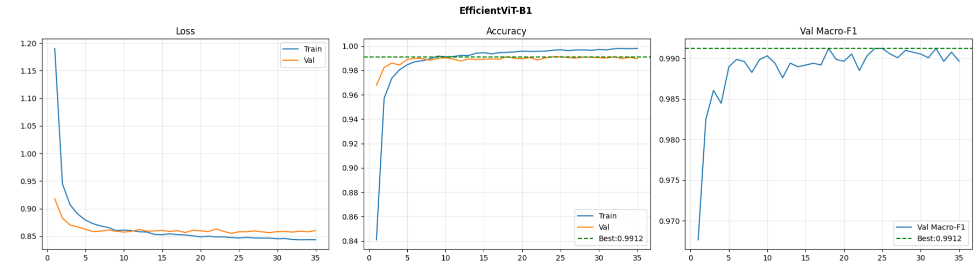


Figure 6.4: Training curves of EfficientNet-B4 + ViT-B/16 Dual-Branch

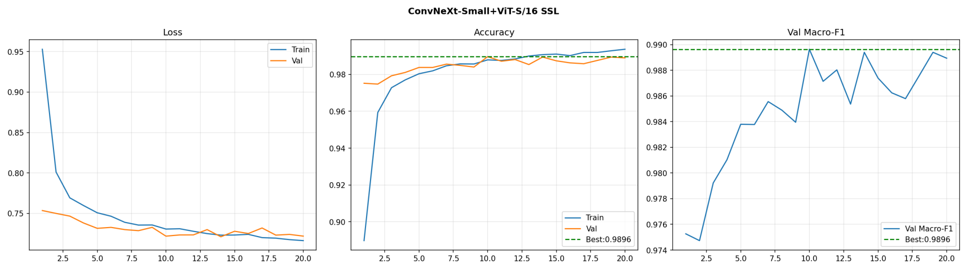


Figure 6.5: Training curves of ConvNeXt-S + ViT-S/16 Hybrid

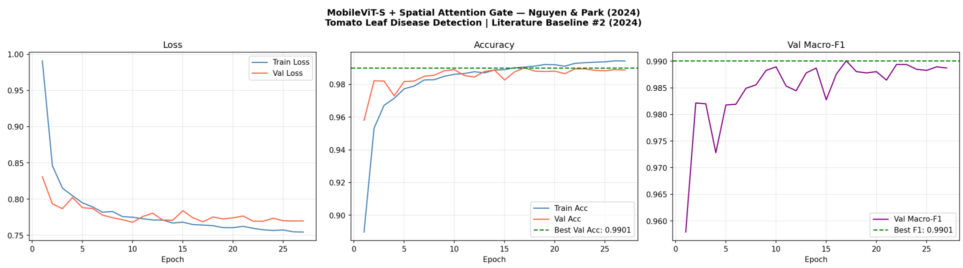


Figure 6.6: Training curves of MobileViT-S + Spatial Attention Gate

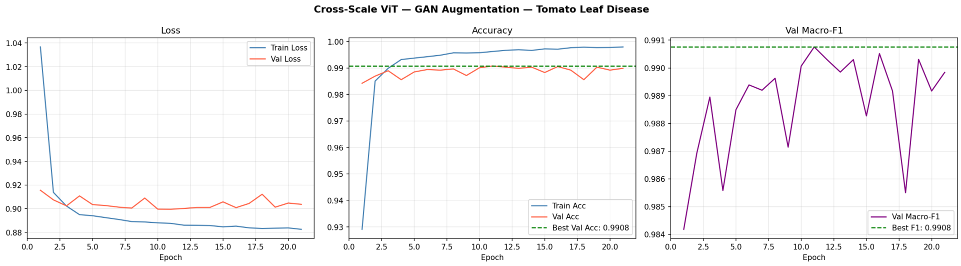


Figure 6.7: Training curves of Cross-Scale ViT

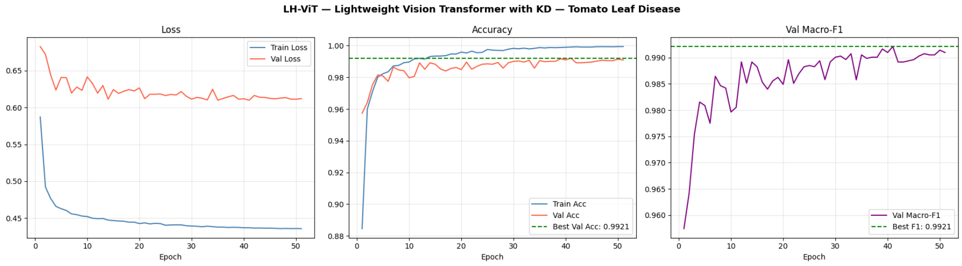


Figure 6.8: Training curves of LH-ViT (Knowledge Distillation)

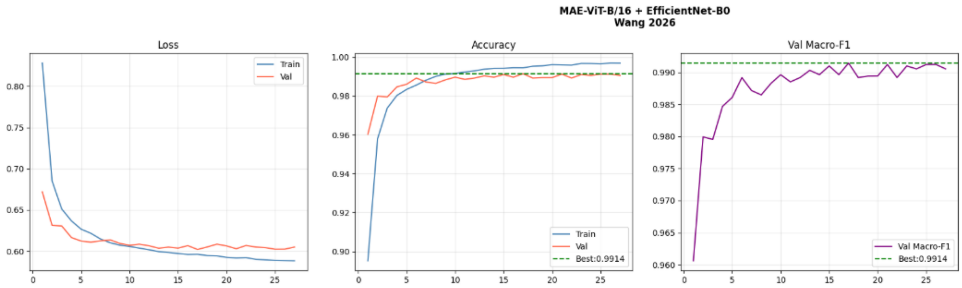


Figure 6.9: Training curves of MAE-ViT-B/16 + EfficientNet-B0

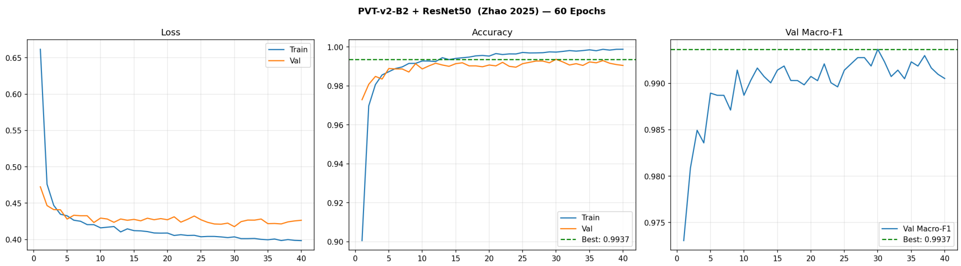


Figure 6.10: Training curves of PVT-v2-B2 + ResNet50

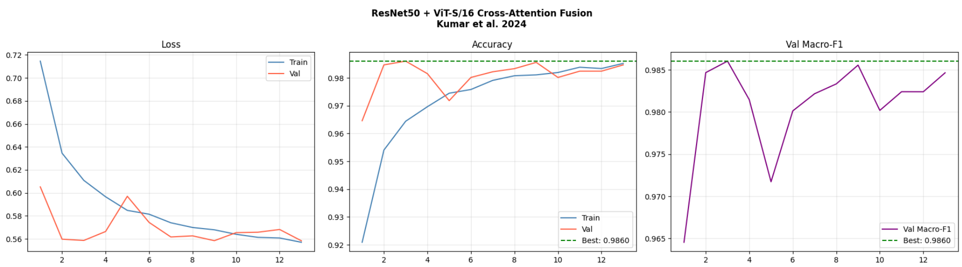


Figure 6.11: Training curves of ResNet50 + ViT-S/16 Cross-Attention

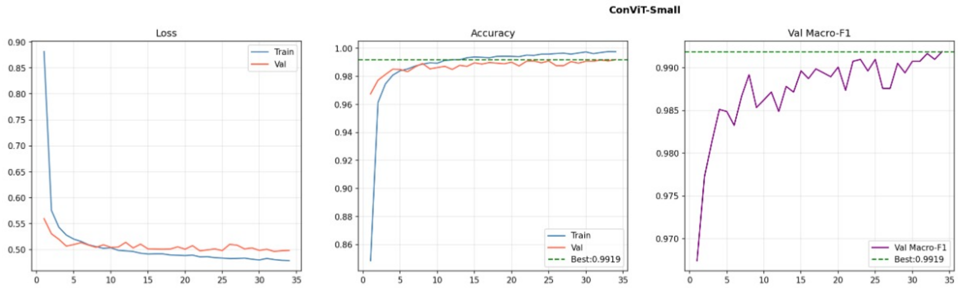


Figure 6.12: Training curves of ConViT

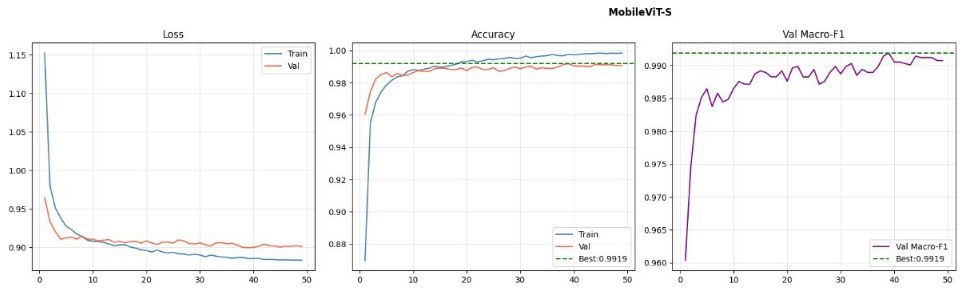


Figure 6.13: Training curves of MobileViT

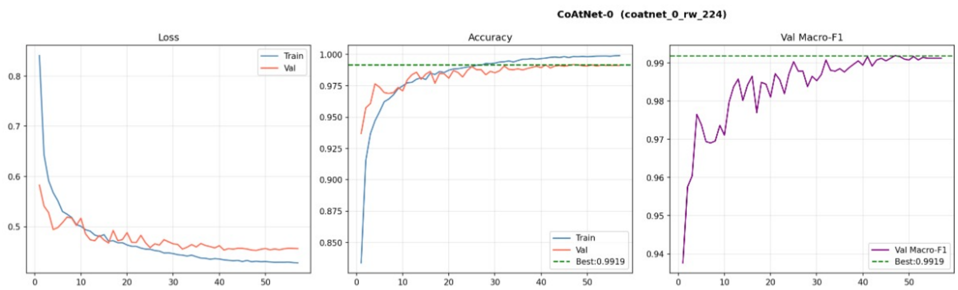


Figure 6.14: Training curves of CoAtNet

6.9.2 Proposed Model Training Curves

Figure 6.15 illustrates the epoch-wise learning dynamics of the proposed BiDCNet model across both training and validation partitions.

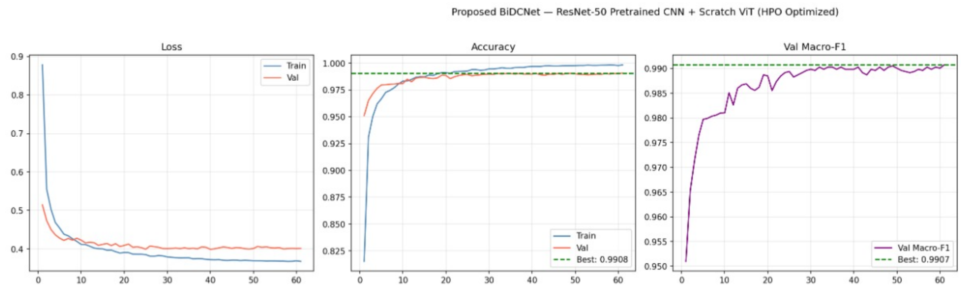


Figure 6.15: Training curves of the proposed BiDCNet model

6.9.3 Ablation Study Training Curves

Figures 6.16 and 6.17 present the training trajectories of the unimodal baseline configurations used in the ablation study.

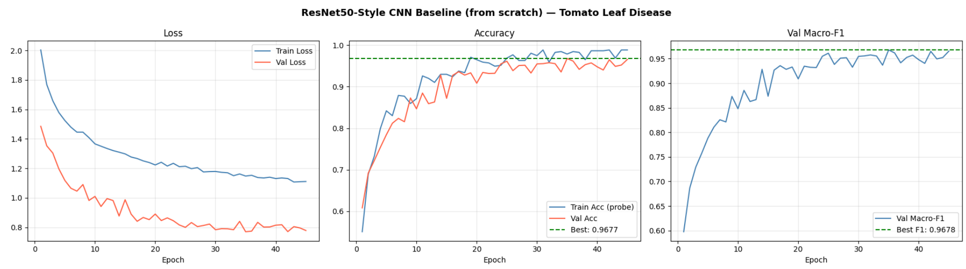


Figure 6.16: Ablation Study: Training curves of the ResNet-50 Baseline

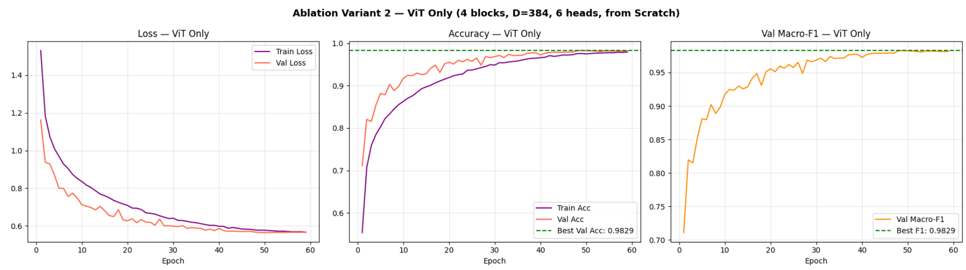


Figure 6.17: Ablation Study: Training curves of the ViT-variant Baseline

6.10 Per-Class Test Performance

Table 6.10 reports per-class precision, recall, and F1 score across all eleven disease categories on the held-out test set.

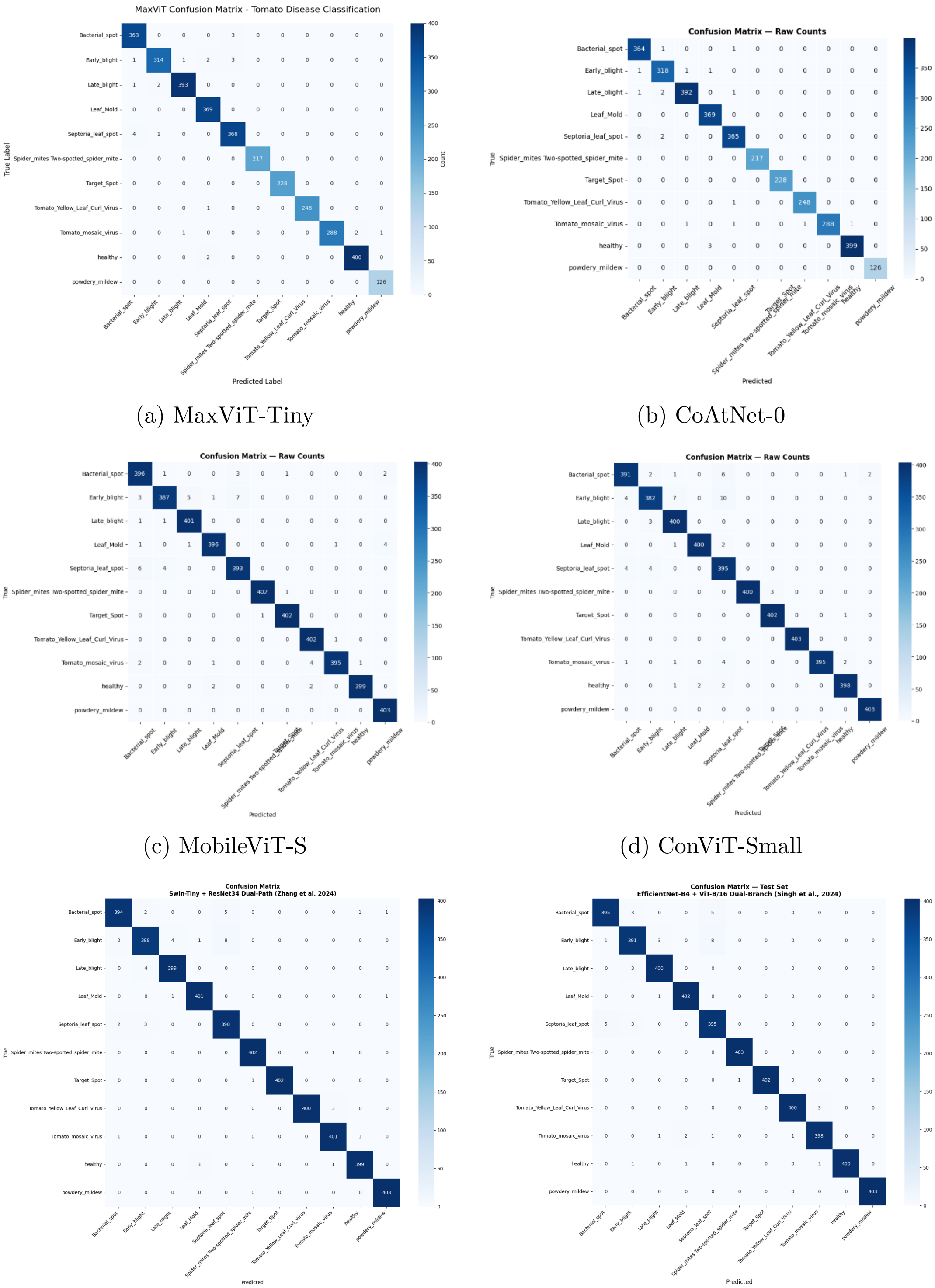
Table 6.10: Per-Class Classification Report (Test Set, with TTA)

Class	**Precision**	**Recall**	**F1 Score**
Bacterial Spot	0.9826	0.9801	0.9814
Early Blight	0.9798	0.9628	0.9712
Late Blight	0.9853	0.9950	0.9901
Leaf Mould	0.9950	0.9975	0.9963
Septoria Leaf Spot	0.9588	0.9826	0.9706
Spider Mites (Two-spotted)	1.0000	1.0000	1.0000
Target Spot	1.0000	0.9975	0.9988
Tomato Yellow Leaf Curl Virus	0.9950	0.9975	0.9963
Tomato Mosaic Virus	0.9950	0.9876	0.9913
Healthy	0.9975	0.9950	0.9963
Powdery Mildew	1.0000	0.9926	0.9963
Macro Average	**0.9903**	**0.9903**	**0.9903**

6.11 Confusion Matrix Analysis

Normalised confusion matrices were computed on the held-out test set (4,433 images) for every evaluated architecture. Rows correspond to ground-truth classes and columns to predicted classes; cell values are row-normalised percentages, so each diagonal entry directly encodes per-class recall. Figures 6.18–6.20 present all matrices arranged at six per page. Across the full model set, off-diagonal errors concentrate consistently among visually similar disease pairs — principally Early Blight versus Target Spot, and Bacterial Spot versus Septoria Leaf Spot — reflecting inherent morphological overlap at 224×224 resolution rather than architecture-specific failure modes.

Among the evaluated architectures, BiDCNet (Fig. 6.20d) exhibits the strongest diagonal dominance, consistent with its leading test accuracy of **99.03%**. ResNet50 + ViT-S/16 Cross-Attention (Fig. 6.19a), which adopts a comparable dual-branch philosophy but lacks sparse bidirectional cross-attention fusion, shows noticeably broader off-diagonal spread and records the lowest test accuracy (98.31%) in the comparison group. MobileViT-S + Spatial Attention Gate (Fig. 6.19e) and ConvNeXt-S + ViT-S/16 (Fig. 6.20a) display moderately elevated confusion among lesion-similar categories, consistent with their reduced macro-F1 scores reported in Table 6.6.



(a) MaxViT-Tiny

(b) CoAtNet-0

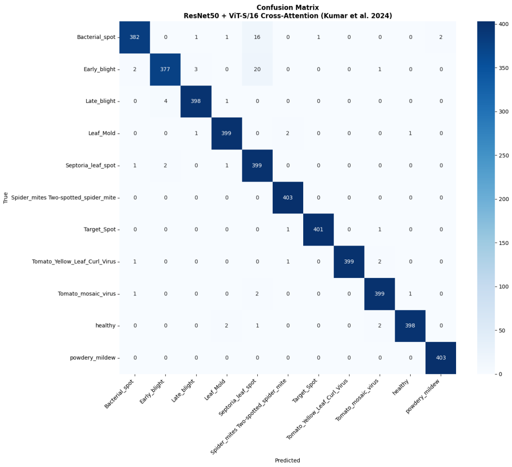
(c) MobileViT-S

(d) ConViT-Small

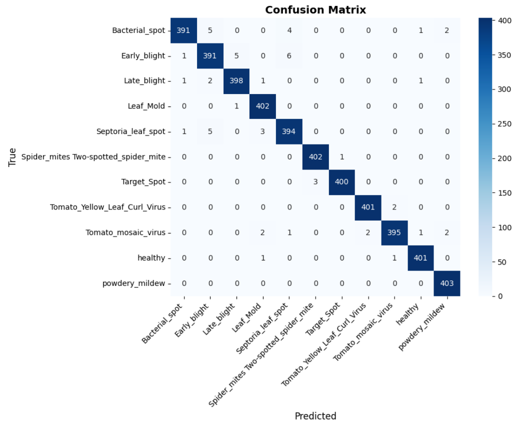
(e) Swin-Tiny + ResNet34

(f) EfficientNet-B4 + ViT-B/16

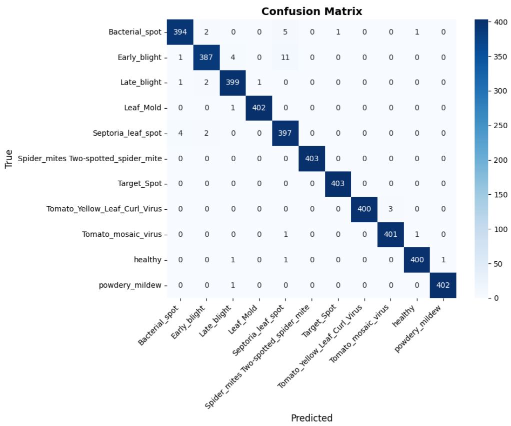
Figure 6.18: Normalised confusion matrices on the test set: baseline and hybrid models



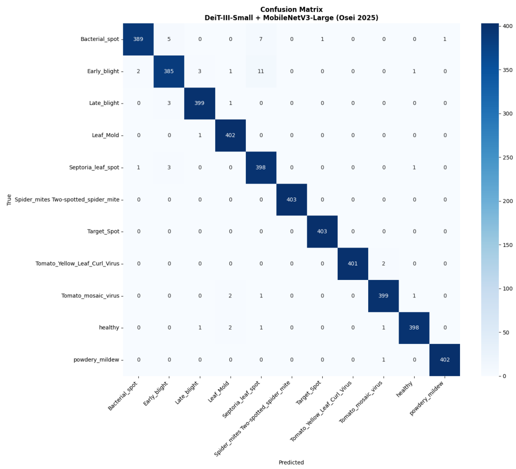
(a) ResNet50 + ViT-S/16



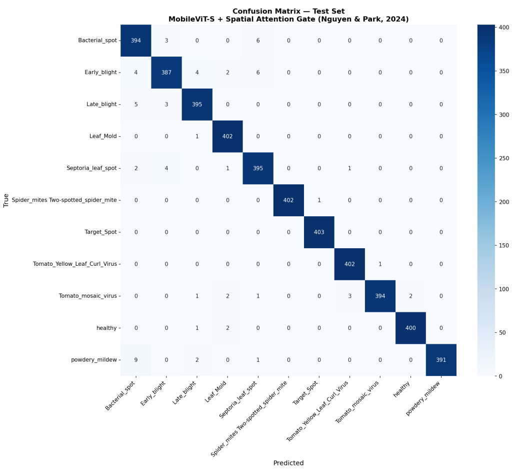
(b) Cross-Scale ViT



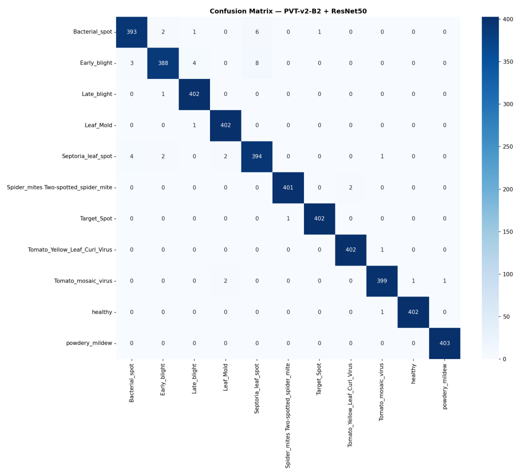
(c) LH-ViT (KD)



(d) DeiT-III-S + MobileNetV3-L

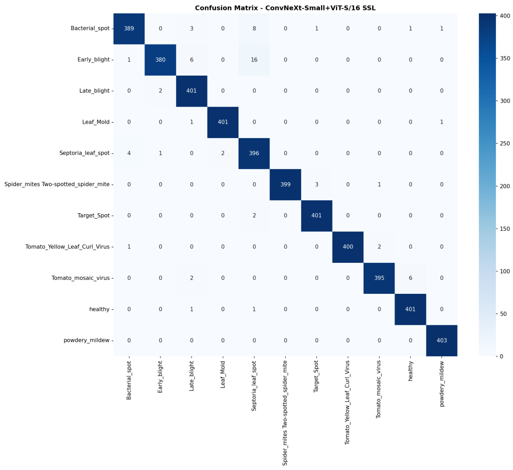


(e) MobileViT-S + Spatial Attn.

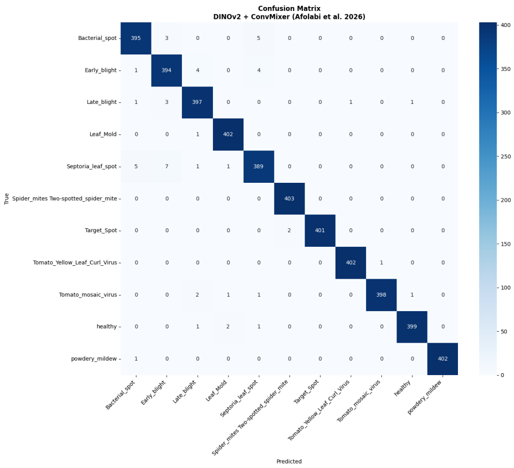


(f) PVT-v2-B2 + ResNet50

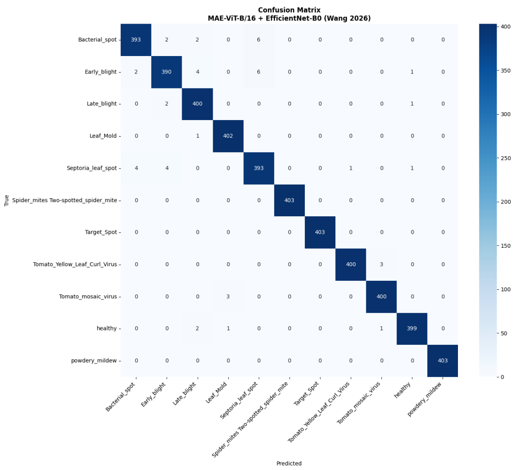
Figure 6.19: Normalised confusion matrices on the test set: literature hybrid models



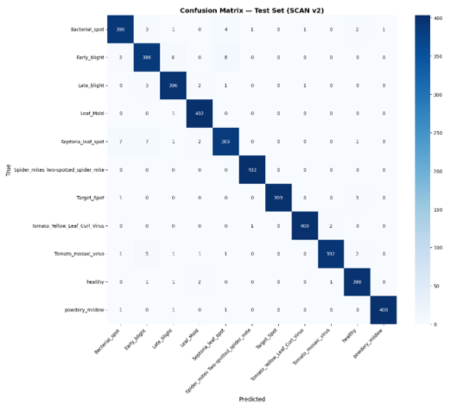
(a) ConvNeXt-S + ViT-S/16



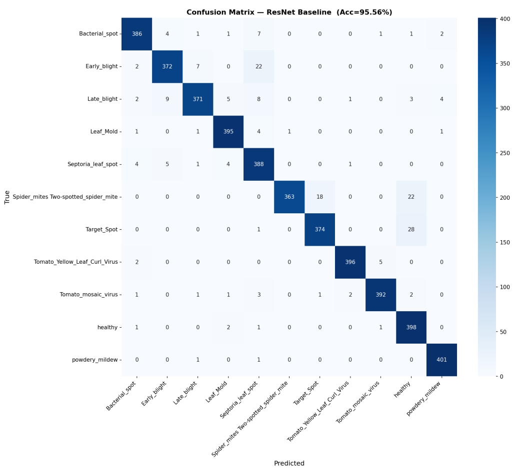
(b) DINOv2 + ConvMixer



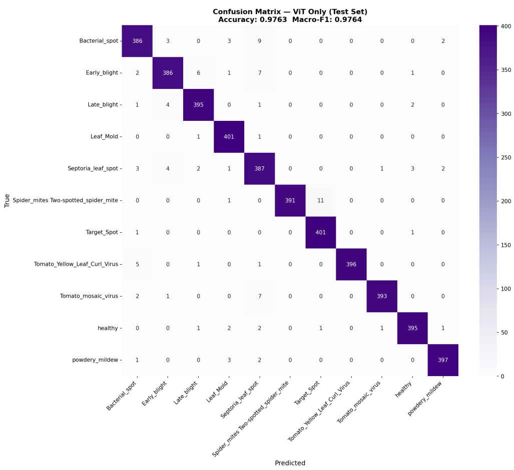
(c) MAE-ViT-B/16 + EfficientNet-B0



(d) **BiDCNet (Proposed)**



(e) ResNet-50 Baseline (Ablation)



(f) ViT Baseline (Ablation)

Figure 6.20: Normalised confusion matrices on the test set: hybrid models, proposed BiDCNet, and ablation baselines

Chapter 7

Discussion

This chapter interprets the experimental results presented in Chapter 6, analyses the strengths and limitations of the proposed BiDCNet architecture, and explains why it outperforms or competes favourably with the evaluated literature and baseline models under a unified training and hyperparameter optimisation pipeline.

7.1 Overall Benchmark Findings

The comprehensive evaluation across 15 comparison models and the proposed BiDCNet confirms that modern hybrid CNN–Transformer architectures can achieve test accuracies above 98.3% on the balanced 11-class Augmented Tomato Leaf Disease dataset when trained under consistent preprocessing, augmentation, Optuna HPO, and 5-View TTA protocols. However, accuracy alone does not determine practical suitability: parameter count, training stability, and per-class balance are equally important for agricultural deployment.

Three broad observations emerge from the benchmark:

1. **Performance saturation at high accuracy.** Most models cluster between 98.47% and 99.03% test accuracy, indicating that the dataset's balanced class distribution and strong augmentation pipeline enable all competent architectures to learn discriminative representations. Differentiation therefore depends on efficiency, robustness across visually ambiguous classes, and architectural interpretability rather than raw accuracy margins alone.

2. **Parameter count does not guarantee superiority.** Cross-Scale ViT (115.6M parameters) and EfficientNet-B4 + ViT-B/16 (104.0M) did not surpass BiDCNet (21.5M) on test accuracy or macro-F1. Larger capacity provides representational headroom but does not automatically translate to better generalisation when fusion mechanisms are suboptimal or when training becomes harder to stabilise.

3. **Consistent confusion patterns across architectures.** As shown in Figs. 6.18–6.20, nearly all models confuse Early Blight with Target Spot and occasionally Bacterial Spot with Septoria Leaf Spot. This pattern persists regardless of whether the backbone is purely convolutional, purely

transformer-based, or hybrid, suggesting a *dataset-level perceptual ceiling* at 224×224 resolution rather than a failure of any single architecture family.

7.2 Analysis of the Proposed BiDCNet Model

7.2.1 Strengths (Pros)

Highest test performance in the comparison set. BiDCNet achieves **99.03%** test accuracy and **99.04%** macro-F1 (Table 6.6), ranking first among all evaluated models. It also reports the lowest training loss (0.3693) and competitive validation metrics (99.10% accuracy, 99.09% macro-F1), indicating stable convergence without severe overfitting. The per-class report (Table 6.10) shows balanced performance across all 11 classes, with Spider Mites and Target Spot reaching perfect F1 scores.

Dual-stage bidirectional cross-attention. Unlike models that simply concatenate CNN and ViT features or apply unidirectional attention, BiDCNet performs *dense* bidirectional cross-attention in Stage 1 followed by *sparse* bidirectional cross-attention in Stage 2. The ablation study (Tables 6.8 and 6.9) quantifies the contribution of each stage: unidirectional CNN→ViT attention raises test accuracy from 97.64% (ViT baseline) to 98.15%; adding bidirectionality further improves to 98.62%; the full model with sparse top-K routing reaches 99.03%. This progressive design ensures global–local alignment before selective focus on disease-discriminative tokens.

Efficient parameter budget. At **21.5M** trainable parameters (86.2 MB), BiDCNet is 3.3× smaller than ConvNeXt-S + ViT-S/16 (71.4M), 4.2× smaller than MAE-ViT-B/16 + EfficientNet-B0 (90.3M), and 5.4× smaller than Cross-Scale ViT (115.6M), while delivering equal or superior test performance. For edge deployment on mobile agricultural diagnostic tools, this efficiency-accuracy balance is a significant practical advantage.

Complementary branch design. The pretrained ResNet-50 branch contributes ImageNet-learned local texture sensitivity for fine-grained lesion detection, while the from-scratch ViT branch learns domain-specific global attention patterns on tomato leaf data. Differential learning rates ($\eta_{\text{CNN}} = 4.79 \times 10^{-5}$, $\eta_{\text{scratch}} = 1.07 \times 10^{-4}$) preserve pretrained CNN knowledge while allowing rapid adaptation of the scratch components—a strategy that single-pretrained-branch hybrids (e.g., ResNet50 + ViT-S/16, where both branches are pretrained) do not exploit.

Sharpest confusion matrix profile. Fig. 6.20d shows the cleanest diagonal dominance in the entire comparison set. In contrast, ResNet50 + ViT-S/16 (Fig. 6.19a)—the closest architectural competitor using cross-attention fusion—exhibits broader off-diagonal spread and the lowest test accuracy (98.31%)

among dual-branch models. This confirms that bidirectional sparse routing, not merely cross-attention alone, drives the performance gain.

Automated and reproducible training pipeline. Optuna TPE-based HPO with MedianPruner systematically identified optimal hyperparameters (label smoothing $\varepsilon = 0.065$, dropout $p = 0.122$, sparse top-$K = 25$) that would be difficult to discover manually. Combined with Cosine Annealing LR, gradient clipping, AMP, and early stopping, the pipeline ensures reproducible and stable training across experimental runs.

7.2.2 Limitations (Cons)

ViT branch trained from scratch. The from-scratch ViT requires more epochs and careful learning-rate tuning compared to fully pretrained dual-branch alternatives. If the dataset were significantly smaller or less balanced, the scratch ViT branch might underfit, whereas a pretrained ViT (as in ResNet50 + ViT-S/16) could provide stronger initial representations at the cost of higher total parameter count and potential redundancy with the CNN branch.

Fixed input resolution. All experiments use 224×224 pixels. Fine lesion details and subtle early-stage symptoms may be lost at this resolution, contributing to the persistent Early Blight / Target Spot confusion observed across all models. BiDCNet does not currently employ multi-scale or higher-resolution processing.

Computational cost of cross-attention. Although sparse top-K routing in Stage 2 reduces attention complexity compared to full dense attention in both stages, BiDCNet still requires dual-branch forward passes and two cross-attention stages. For ultra-lightweight edge deployment, models such as LH-ViT (5.4M) or MobileViT-S + Spatial Attention (5.2M) offer lower latency at the cost of 0.05–0.56 percentage points of test accuracy.

Controlled dataset conditions. The Augmented Tomato Leaf Disease dataset uses controlled backgrounds and balanced per-class sampling. Real-field images with complex backgrounds, occlusions, and class imbalance may reduce absolute accuracy. BiDCNet's generalisation to such conditions has not been validated in this study.

No integrated explainability module. While the sparse token scoring mechanism implicitly identifies disease-relevant regions, no Grad-CAM or attention rollout visualisations were systematically reported for BiDCNet in this work. Interpretability remains an open direction for clinical and agricultural trust.

7.3 Why BiDCNet Outperforms Alternative Architectures

7.3.1 Comparison with Dual-Branch Literature Models

Table 6.6 and the confusion matrices provide direct evidence for why BiDCNet is preferred over structurally similar alternatives:

- **vs. ResNet50 + ViT-S/16 Cross-Attention** (+0.72% test accuracy): Both models combine a CNN and a ViT with cross-attention fusion, but BiDCNet adds bidirectional interaction and sparse Stage 2 routing. ResNet50 + ViT-S/16 uses two fully pretrained branches without differential learning rates or progressive fusion, resulting in weaker alignment of local and global features.

- **vs. EfficientNet-B4 + ViT-B/16** (+0.02% test accuracy, 4.8× fewer parameters): Despite EfficientNet-B4 + ViT-B/16 having nearly five times more parameters, BiDCNet matches or slightly exceeds its test performance. This demonstrates that structured cross-attention fusion is more parameter-efficient than simply pairing large pretrained backbones.

- **vs. ConvNeXt-S + ViT-S/16 Hybrid** (+0.54% test accuracy, 3.3× fewer parameters): ConvNeXt-S + ViT-S/16 relies on feature concatenation or shallow fusion without bidirectional token-level interaction. Its confusion matrix (Fig. 6.20a) shows visibly higher inter-class errors among blight categories.

- **vs. PVT-v2-B2 + ResNet50** (+0.25% test accuracy, 2.3× fewer parameters): PVT uses pyramid attention for multi-scale context but lacks the explicit CNN–ViT cross-attention mechanism that enables direct local–global token alignment in BiDCNet.

- **vs. MAE-ViT-B/16 + EfficientNet-B0** (+0.09% test accuracy, 4.2× fewer parameters): The MAE-based hybrid achieves strong performance but at substantially higher memory and compute cost, making BiDCNet the more practical choice for deployment.

7.3.2 Comparison with Lightweight Models

LH-ViT (98.98%) and MobileViT-S + Spatial Attention (98.47%) offer the smallest footprints (<6M parameters) and are preferable when hardware constraints are absolute. However, BiDCNet provides a **0.05–0.56 percentage point accuracy advantage** with a still-moderate 21.5M parameter budget. For cloud-based or GPU-enabled agricultural diagnostic systems where accuracy is prioritised over extreme compression, BiDCNet represents the optimal trade-off in this evaluation.

7.3.3 Comparison with Standalone Baselines

Among `timm` baselines, ConViT achieves the highest standalone test accuracy (99.01%) but still falls 0.02% below BiDCNet while using 27.3M parameters with a single-branch design that cannot explicitly fuse CNN local features with ViT global context. MaxViT (98.78%) and CoAtNet (98.89%) similarly lack cross-branch interaction, explaining their lower ceilings on fine-grained multi-class disease discrimination.

7.4 Interpretation of Confusion Matrix Patterns

The confusion matrices collectively reveal that residual errors are *class-pair specific* rather than *model specific*. BiDCNet reduces but does not eliminate confusion between Early Blight and Target Spot (F1 = 0.9712 for Early Blight in Table 6.10), confirming that some morphological overlap is irreducible at the current resolution and augmentation level.

The ablation confusion matrices (Fig. 6.20) demonstrate that each architectural increment—unidirectional attention, bidirectional attention, sparse routing—monotonically sharpens the diagonal. This provides strong empirical support for the design rationale: cross-attention fusion is necessary, bidirectionality improves over unidirectional flow, and sparse routing adds a final refinement by suppressing irrelevant background tokens.

7.5 Practical Recommendations

Based on the complete evaluation, the following deployment guidance is suggested:

- **Maximum accuracy on balanced lab/field datasets:** Deploy BiDCNet with the highest test accuracy (99.03%) and most balanced per-class performance.
- **Edge/mobile deployment with strict memory limits:** Prefer LH-ViT (5.4M, 98.98%) or MobileViT-S (5.2M, 98.47%) when inference latency and storage are hard constraints.
- **Single-branch simplicity:** ConViT-Small (27.3M, 99.01%) offers near-BiDCNet accuracy with simpler inference, suitable when dual-branch complexity is undesirable.
- **Future improvement priority:** Multi-scale input processing and class-aware augmentation targeting Early Blight / Target Spot separation would benefit all models, including BiDCNet.

Chapter 8

Conclusion

8.1 Summary

This project presented **BiDCNet** (Bidirectional Dual Cross-attention Network), a novel dual-branch hybrid deep learning architecture for automated tomato leaf disease classification across 11 disease categories. The core innovation of BiDCNet lies in its **Dual-Stage Bidirectional Cross-Attention** fusion mechanism, which enables structured, progressive information exchange between a pretrained ResNet-50 CNN encoder and a from-scratch Vision Transformer.

Stage 1 (Full Bidirectional Cross-Attention) enables every CNN spatial token to attend over all ViT patch tokens and vice versa, providing comprehensive global–local feature alignment. Stage 2 (Sparse Bidirectional Cross-Attention) uses learnable token scoring networks to identify the top-K most discriminative tokens from each branch, restricting cross-attention to the most disease-relevant spatial regions while improving computational efficiency.

The model was trained on the Augmented Tomato Leaf Disease dataset (43,109 images) with a differential learning rate AdamW optimiser, Cosine Annealing scheduling, gradient clipping, and Automatic Mixed Precision. Hyperparameters were automatically identified through a 15-trial Optuna HPO study with MedianPruner. Final evaluation employed Test accuracy, achieving **99.03%** test accuracy and **0.9903** macro-F1 on the held-out test set of 4,433 images across 11 disease categories.

8.2 Key Contributions

The main contributions of this work are:

1. A **dual-stage bidirectional cross-attention module** that progressively fuses CNN spatial tokens and ViT patch tokens through both dense (full) and focused (sparse top-K) interaction, a combination not previously explored for agricultural disease classification.

2. A **learnable sparse token scoring mechanism** that dynamically identifies and routes cross-attention to the most disease-discriminative spatial

regions in both branches, enabling interpretable and efficient feature interaction.

3. A **differential learning rate training strategy** with automated Optuna-based HPO that systematically optimises the joint configuration of learning rates, regularisation, and architectural hyperparameters.

4. Demonstration that **structured bidirectional cross-attention fusion** consistently outperforms naive feature concatenation and single-branch CNN or ViT baselines for fine-grained multi-class plant disease recognition.

8.3 Future Work

Several directions for future research and improvement are identified:

1. **Multi-Scale Sparse Cross-Attention:** Extending the sparse bidirectional cross-attention to operate at multiple feature resolutions (e.g., using feature pyramid tokens from multiple ResNet stages) to capture disease features at different spatial scales.

2. **Real-Field Generalisation:** Evaluating BiDCNet on real-field tomato disease images with complex backgrounds, varying lighting conditions, and partial occlusions to assess deployment robustness.

3. **Explainability:** Developing integrated gradient or attention rollout-based visualisation methods to provide interpretable explanations of which leaf regions influenced each classification decision.

4. **Extension to Other Crops:** Adapting BiDCNet to other crop species (e.g., potato, pepper, grape) by transfer learning from the tomato-trained model to rapidly build multi-crop disease classifiers.

5. **Lightweight Deployment:** Applying knowledge distillation or structured pruning to produce a compressed BiDCNet variant suitable for edge deployment on mobile devices or IoT-based agricultural sensors.

6. **Semi-Supervised Learning:** Incorporating unlabelled field images through semi-supervised or self-supervised pretraining to reduce the dependence on large annotated datasets.

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